Bogdanovic Stefan, Rodari Simone,
Pignataro Gabirele

POLITECNICO
MILANO 1863

Students&Companies
Requirement Analysis and Specification Document

Deliverable:	RASD
Title:	Students&Companies
Authors:	Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele
Version:	1.0
Date:	22-December-2024
Download page:	https://github.com/SimoPolimi/RodariPignataroBogdanovic
Copyright:	Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Contents

Table of Contents 3

List of Figures 5

List of Tables 5

1 Introduction 6
- 1.1 Purpose 6
  - 1.1.1 Goals 6
- 1.2 Scope 7
  - 1.2.1 World Phenomena 7
  - 1.2.2 Shared Phenomena 7
- 1.3 Definitions, Acronyms, Abbreviations 8
  - 1.3.1 Definitions 8
  - 1.3.2 Acronyms 8
  - 1.3.3 Abbreviations 8
- 1.4 Revision History 8
- 1.5 Reference Documents 8
- 1.6 Document Structure 9

2 Overall Description 10
- 2.1 Product Perspective 10
  - 2.1.1 Scenarios 10
  - 2.1.2 Domain Class Diagram 12
  - 2.1.3 State Diagrams 13
- 2.2 Product Functions 13
  - 2.2.1 Student 13
  - 2.2.2 Company 14
  - 2.2.3 University 14
  - 2.2.4 Students and Companies 14
  - 2.2.5 Everyone 14
- 2.3 User Characteristic 15
  - 2.3.1 Student 15
  - 2.3.2 Company 15
  - 2.3.3 University 15
- 2.4 Assumptions, Dependencies and Constraints 15
  - 2.4.1 Regulatory Policies 15
  - 2.4.2 Domain Assumptions 15

3 Specific Requirements 16
- 3.1 External Interface Requirements 16
  - 3.1.1 User Interfaces 16
  - 3.1.2 Hardware Interfaces 20
  - 3.1.3 Software Interfaces 20
  - 3.1.4 Communication Interfaces 20
- 3.2 Functional Requirements 20
  - 3.2.1 Student 20
  - 3.2.2 Company 21
  - 3.2.3 University 21

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

3.2.4 Students and Companies . . . 22
3.2.5 Everyone . . . 22
3.2.6 Use Cases Diagram . . . 23
3.2.7 Use Cases . . . 24
3.2.8 Sequence Diagrams . . . 32
3.2.9 Requirement Mapping . . . 41
3.3 Performance Requirements . . . 51
3.4 Design Constraints . . . 51
3.4.1 Standards Compliance . . . 51
3.4.2 Hardware and Software Limitations . . . 52
3.5 Software System Attributes . . . 52
3.5.1 Availability . . . 52
3.5.2 Security . . . 52
3.5.3 Maintainability . . . 52
3.5.4 Portability . . . 52

4 Formal Analysis Using Alloy . . . 53
4.1 Examples . . . 58

5 Effort Spent . . . 62

References . . . 63

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

List of Figures

1 Domain class diagram 12
2 Contract lifecycle state diagram 13
3 User Login 16
4 Edit Profile 17
5 Search Internship 17
6 Create Questionnaire 18
7 Discussion 18
8 Upload Internship 19
9 Monitor Status 19
10 ShowSingle, first two steps 58
11 ShowSingle, first two steps 58
12 Show, first two steps 59
13 Show, last two steps 59
14 Show with Feedback, last two steps 60
15 ShowInterrupted, last two steps 60
16 ShowWithSkills, first two steps 61

List of Tables

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

1 Introduction

1.1 Purpose

Internships are a vital bridge between academic learning and professional experience, providing students with opportunities to apply their skills and companies with access to emerging talent. However, finding the right match between students and internships can be a complex and inefficient process. The Students&Companies (S&C) platform addresses these challenges by providing an integrated system that connects university students with companies offering internships. S&C provides a wide range of features offering a seamless experience for students from the search of an internship to its completion and for companies an easy way to advertise their internship projects. The platform also lets universities to monitor and support their students throughout their internship, overseeing its execution and supporting the involved parties, should problems arise.

1.1.1 Goals

[G1] Students must be able to search for internships posted by companies.

[G2] Companies must be able to upload their internships offers.

[G3] Students must be notified when an internship that could interest them becomes available.

[G4] Companies must be notified about available students with CVs and skills matching their internship requirements.

[G5] Students must be able to apply for internships of interest.

[G6] Companies must be able to accept applicants to their internship.

[G7] Companies must be supported in conducting a selection process for applicants, including the collection of information about the students using questionnaires and the management of interviews.

[G8] Students want to receive suggestions on how to make their CVs more appealing.

[G9] Companies want to receive suggestions on how to make their project descriptions more appealing.

[G10.A] Students must be able to keep track and monitor the execution and the outcomes of the matchmaking process.

[G10.B] Companies must be able to keep track and monitor the execution and the outcomes of the matchmaking process.

[G10.C] Universities must be able to monitor the status of the internship in which its students have applied.

[G11] The interested parties of an internship (students, companies and universities) must be able to communicate with each other about info and eventual problems regarding the ongoing internship.

[G12.A] Companies must be able to file complaints about students interning with them.

[G12.B] Students must be able to file complaints about the companies where they are interning.

[G13] Universities must be able to review complaints about their students and the companies where their students are interning.

[G14] Universities must be able to interrupt internships between their students and a company.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

1.2 Scope

The Students&Companies platform provides its services to different users, identifiable into three main classes: students, companies and universities. Each one of them has different roles and interacts with the system in various ways. S&C is aimed mainly at university students that want to take part in an internship program, as they can search for new internships suiting their preferences or let the system inform them when internships matching their profile are available. S&C is also able to provide suggestions on how to improve their profile to make it more appealing to companies. The platform also supports companies easing the internships' management. In doing so S&C can be used to create questionnaires to select the most suitable candidates for an internship. Similarly to the suggestions for students, the system can analyze the internship offer to identify eventual improvements that can be made. Hopefully, every internship comes to an end without problems but if that is not the case, interns and companies can communicate through the platform possibly involving the intern's alma mater as a supervisor which can also terminate the internship if the parties cannot resolve the controversy.

1.2.1 World Phenomena

[WP1] Students want to do an internship.

[WP2] Companies offer an internship.

1.2.2 Shared Phenomena

World Controlled

[SP1] Companies upload their internship offers.

[SP2] Students search for an internship.

[SP3] Students accept the internship.

[SP4] Companies approve the internship contract for a student.

[SP5] Companies interview the student.

[SP6] Students give feedback on the internship.

[SP7] Companies give feedback on the internship.

[SP8] Students file complaints about the internship.

[SP9] Companies file complaints about the internship.

[SP10] University monitors the situation of internships.

[SP11] University receive a complaint.

[SP12] University responds to a complaint.

[SP13] University terminates an internship.

[SP14] Students keeps track, monitors the execution and the outcomes of the matchmaking process.

[SP15] Companies keeps track, monitors the execution and the outcomes of the matchmaking process.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Machine Controlled

[SP16] System informs the student when an internship that might interest him becomes available.

[SP17] System informs the companies about the availability of student CV corresponding to their needs.

[SP18] System establishes a contact between the student and the companies.

[SP19] System analyzes internship data for recommendations.

[SP20] System sends suggestions on how to improve the students CVs.

[SP21] System sends suggestions on how to improve companies project descriptions.

[SP22] System sends universities complaints.

[SP23] System sends a notification to the university about the complaint.

1.3 Definitions, Acronyms, Abbreviations

1.3.1 Definitions

- **Recommendations**: Process in which the system informs students when an internship that might interest them becomes available and can inform companies about the availability of student CVs corresponding to their needs.
- **SheerID**[2]: Platform that offers an API to verify students by using their academic credentials.
- **Contract**: The contract represents the lifecycle of an internship application, from initiation to its conclusion.

1.3.2 Acronyms

- **S&C**: Students & Companies

1.3.3 Abbreviations

- **G***: goals.
- **WP***: world phenomena.
- **SP***: shared phenomena.
- **R***: functional requirements.
- **UC***: use cases.

1.4 Revision History

- 22/12/2024 - Version 1.0.
- 07/01/2025 - Version 2.0.

1.5 Reference Documents

- Assignment document “Assignment RDD AY 2024-2025.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

1.6 Document Structure

Introduction: This section provides a concise overview of the project, focusing on the objectives and the reasons for its development. It outlines the key goals that the project aims to achieve. **Overall description**: This section offers a high-level explanation of how the system operates, providing a detailed account of the phenomena involved in the interaction between the world, the machine, or both. It also includes a description of the domain, along with the assumptions that underpin the system's design and functionality.
Specific requirements: Here, a thorough analysis of the requirements necessary to meet the project's goals is provided. This section also includes important information for developers, such as hardware and software constraints, along with any other specifications that guide the development process.
Formal analysis using Alloy: This section presents a formal representation of world phenomena, using the Alloy language to describe the system's behavior and properties in a precise and logical manner.
Effort spent: This section details the time and resources invested in the creation of the document, broken down by each section for clarity and transparency. References: This section includes citations and references to any documents, frameworks, or software tools utilized in the development of this paper.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

2 Overall Description

2.1 Product Perspective

2.1.1 Scenarios

Student edits his profile
A student decides to enhance their chances of finding an internship by creating a detailed profile. They access using their academic credentials and start the profile editing. They upload a CV to make it available to companies and list their skills, which will be used by the platform to find the best matching internship. After submitting the changes, the system analyzes it and suggests improvements, such as adding missing skills or refining the description of experiences. At any time the student can update his profile.

Student searches and applies for an internship
A student is eager to find an internship aligned with their career aspirations. He explores the available opportunities by filtering results based on their interests and skill requirements. The results of these operations are ordered based on the match between the requested skills and the ones specified in his profile. Once they find a suitable internship, they review the project details carefully and submit their application. The system forwards their CV to the company for evaluation.

The system recommends internships to a student
After completing their profile and specifying their skills and preferences, such as desired fields or specific technologies, a student receives notifications from the system whenever an internship aligns with their profile. The student reviews these recommendations and selects the most relevant opportunities to apply for.

Company posts an internship project
A company wants to post an internship project on the platform in order to attract qualified candidates. To do this it logs in and from its dashboard selects the section “Create a new internship”. From this section the company sets the following info about the project:

- A (unique) title to identify the project.
- A description to explain in more detail what the internship is about.
- The terms of the internship, such as the retribution or the work hours.
- A set of tags that specify which skills are better suited for the offer.

Once this step is completed, the user can see the posting from a dedicated section in the dashboard.

The system recommends candidates to companies
After companies post internship opportunities, the system analyzes the profiles of students who have expressed interest in internships. Based on the skills, qualifications, and preferences of both the students and the internships, the system sends notifications to companies when candidates who match their requirements are available. Companies can then review the recommended student profiles and decide whether to proceed with the selection process. Once a company selects a candidate from the recommended list, the system notifies the student and updates the status of the internship accordingly. The company can then initiate the selection process, which may involve interviews or assessments, to finalize the decision.

A company receives an application and starts an interview

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Once a student applies for an internship, the company is notified and begins the selection process. To assess if the candidate is a good fit, the company reviews the candidate's profile, including the CV. For a better evaluation of aspects that may not be covered by the first step, the company can create questionnaires through a dedicated section in its profile. The candidate has then to answer to them through the platform. These interviews may involve structured questionnaires or discussions aimed at gauging the candidate's skills, motivation, and suitability for the role. The system supports the company by helping schedule interviews, managing the process, and collecting answers from the student. It also stores feedback and assessment results to assist the company in making an informed decision. After the interviews, the company determines whether to proceed with the candidate and updates the status of the internship accordingly. The student is then notified about the outcome and provided with next steps for finalizing the internship agreement.

A company accepts a candidate for an internship
After evaluating applications and conducting interviews or assessments, a company identifies the most suitable candidate for an internship position. The company confirms their choice by notifying the candidate through the platform. The system updates the status of the internship to reflect the selection, and the student receives a confirmation along with any necessary next steps to finalize the agreement.

Improving student profiles and CVs
The system reviews student profiles and identifies opportunities for improvement. It suggests refining CV descriptions, highlighting key achievements, or adding new skills to align better with the needs of prospective internship postings. These recommendations help students present themselves as stronger candidates.

A student posts feedback about an internship
A student has ended an internship and since he had a very positive experience, he decides to leave some feedback to let others know about it. He can add a rating from 1 to 5 stars and, optionally, leave a textual comment where he can explain in more detail the reasons for his rating.

Optimizing internship descriptions
The system evaluates internship postings to ensure they are clear and appealing to potential applicants. It might recommend refining project details, adjusting skill requirements to better match available candidates, or emphasizing benefits such as mentorship opportunities or financial compensation. These enhancements increase the visibility and attractiveness of the internship.

A company leaves feedback about an internship
After the completion of an internship, a company that has worked with a student decides to leave feedback to let others know about the performance. The company can add a rating from 1 to 5 stars and, optionally, leave a textual comment where he can explain in more detail the reasons for his rating.

Handling a complaint during an internship
During an internship, issues may arise, such as unclear project requirements, insufficient support from the company, or performance concerns from either the student or the company. Both parties, students and companies, can report problems through the platform. The system ensures that complaints are properly recorded and escalated to the relevant university representatives for resolution. If necessary, the internship is paused or terminated, and both the student and the company are notified. The university then coordinates the resolution process, ensuring that any serious concerns are addressed promptly.

Monitoring an ongoing internship
Relevant parties, such as the company hosting the internship, the student participating in it, and the uni-

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

versity overseeing the program, may have an interest in actively monitoring the internship's progress, status, and outcomes to ensure alignment with expectations, compliance with agreed-upon goals, and the overall success of the experience.

2.1.2 Domain Class Diagram

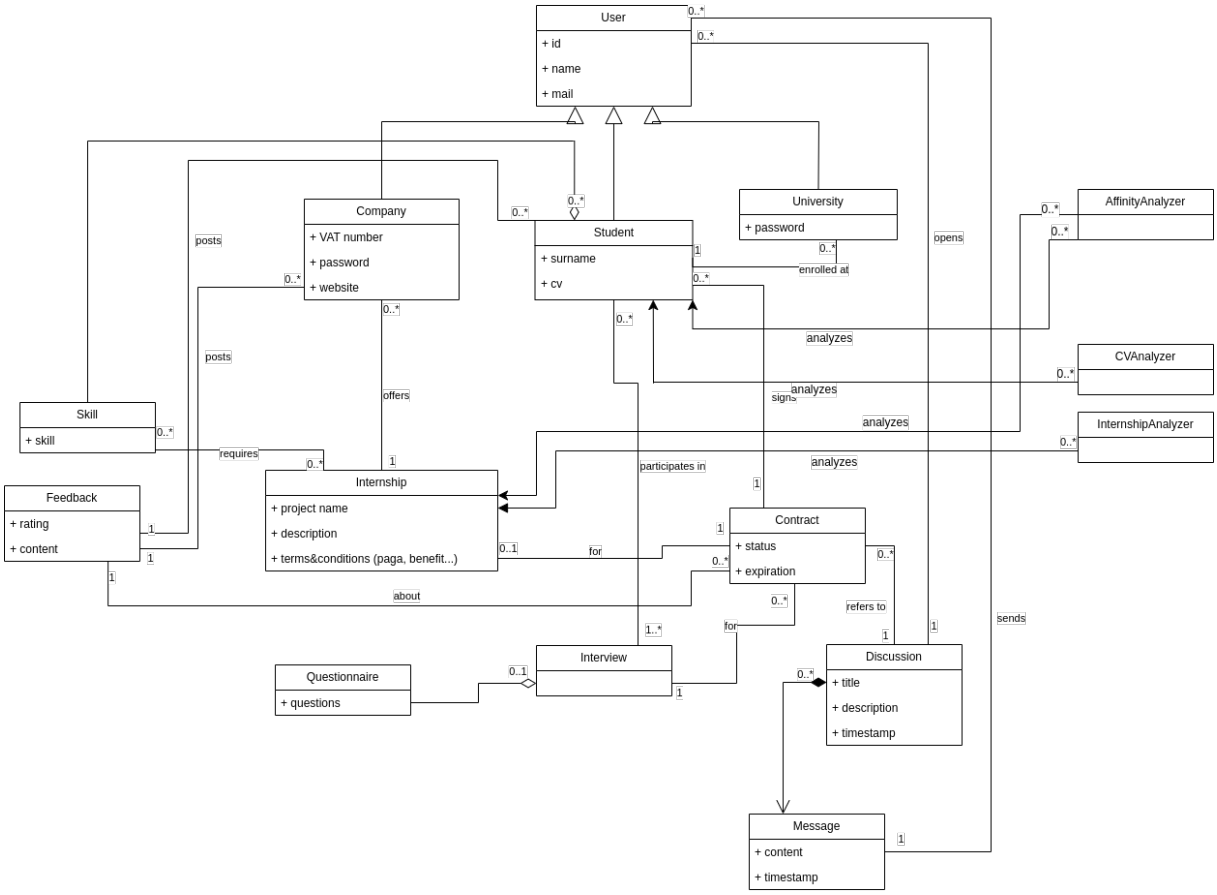


Figure 1: Domain class diagram

The figure above presents the domain class diagram. Company, Student, and University represent distinct user types, each a subclass of an abstract User class. A company offers internships to which multiple students may apply. Each interview is associated with a single student and internship, encompassing multiple questions. Following an interview, a contract, referencing a single internship and student, is generated; its status is tracked via the associated InternshipStatus.

To register a complaint, report an issue, or submit information, users must initiate a discussion thread to facilitate communication among relevant stakeholders. Upon resolution, the initiating user will close the thread and mark it as resolved.

Should a consensus remain elusive, the university may terminate the internship by modifying the student's enrollment status.

The Affinity Analyzer quantifies the compatibility between a student and an internship, considering the student's CV, internship specifications, and prior feedback from both student and company regarding other internships.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

The CVAnalyzer offers recommendations for CV enhancement based on a comparative analysis of other CVs and internships.

The InternshipAnalyzer provides feedback on optimizing internship descriptions by referencing other internships and student CVs.

2.1.3 State Diagrams

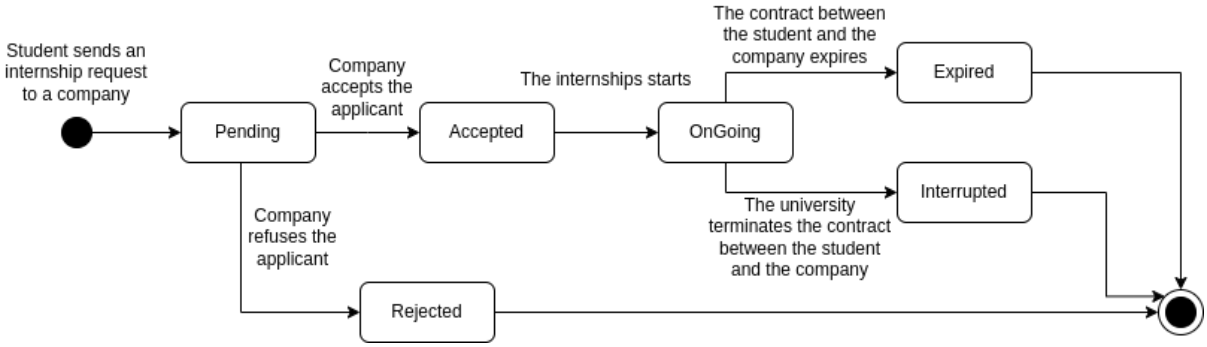


Figure 2: Contract lifecycle state diagram

One of the most important aspect of Students&Companies is the internships' management. In 2 are illustrated the steps followed by the contracts. Note that *Pending* means that the student has applied for an internship and if the company accepts their application, possibly after a round on interviews, the internship can start.

2.2 Product Functions

In this section the functions provided by the system are described to be categorized by user.

2.2.1 Student

Login
The system allows students to log on to the platform using their academic credentials.

Edit Profile
The system allows students to edit their profile.

Internship recommendation
This function allows students to receive an email notification when an internship that could interest them becomes available.

Search an internships
This function enables students to search for internships using keyword filters.

Apply for an internship
This function allows the student to apply for an internship.

Answer interview questions
This function enables students contacted by the company to answer structured questionnaires.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Consulting CV suggestions
Students can view system-generated suggestions on how to improve their CVs to make them more attractive to companies.

2.2.2 Company

Registration and login The application allows companies to register and become users. Registration requires the company to provide its name and tax identification number (P. IVA number), [altre informazioni?]. This information is mandatory; incomplete submissions will result in registration failure.

Upload an internship
This function allows logged-in companies to create new internship posts. They do this by inserting mandatory data, including the internship type, duration, and whether it is paid, along with optional information.

Student recommendation
This function allows companies to receive an email notification when a student CVs corresponding to their needs becomes available.

Interview students
This function allows companies to interview students by proposing structured questionnaires.

Create contract
This function allows companies to register the start of an internship by saving the contract signed by both parties and other optional information.

2.2.3 University

Login
The system allows universities to log into the platform using the credentials given by S&C at the stipulation of the partnership.

Terminate an internship
This function allows university supervisors to terminate an internship by changing its status to "terminated" when an agreement cannot be reached.

2.2.4 Students and Companies

Give feedback about an internship
This feature allows both students and companies to post feedback about internships, which includes a required rating and an optional textual comment.

Open discussion
Users may initiate discussions with interested parties.

2.2.5 Everyone

Send and receive messages within a discussion
This function allows the user involved in a discussion to exchange messages with other interested parties.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Monitor the status of the internship
This function allows the user to see the status of an internship in which he's involved.

2.3 User Characteristic

There are three types of users that interact with the Platform: student, company and university.

2.3.1 Student

University students seeking internships. They can actively search for internships or receive automated recommendations.

2.3.2 Company

Company is a user who can create an internship and interview students with structured questionnaires. This type of user can also start a discussion.

2.3.3 University

University is a user who can interact with students and companies in a discussion and end an internship when needed.

2.4 Assumptions, Dependencies and Constraints

2.4.1 Regulatory Policies

S&C complies with GDPR, CCPA, and other applicable data protection laws. Personal data (students, companies, universities) is collected and processed transparently, only for purposes such as managing internships.

2.4.2 Domain Assumptions

[D1] Students create their CVs.

[D2] Companies properly insert information about internships.

[D3] Universities interfere with the contract between students and companies when needed.

[D4] Notifications must arrive to the target users.

[D5] Students have an academic credential.

[D6] University is available on the SheerID API.

[D7] Students are enrolled in a university registered on SheerID.

[D8] Users who register as a company have a valid VAT number.

[D9] Users who register as a company have a valid VAT number.

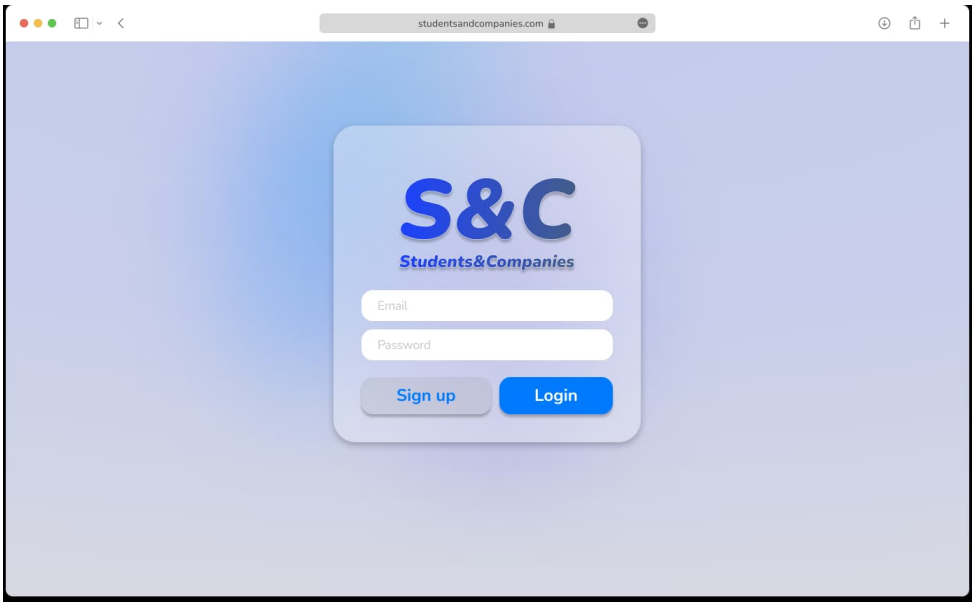
Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

Figure 3: User Login



Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Figure 4: Edit Profile

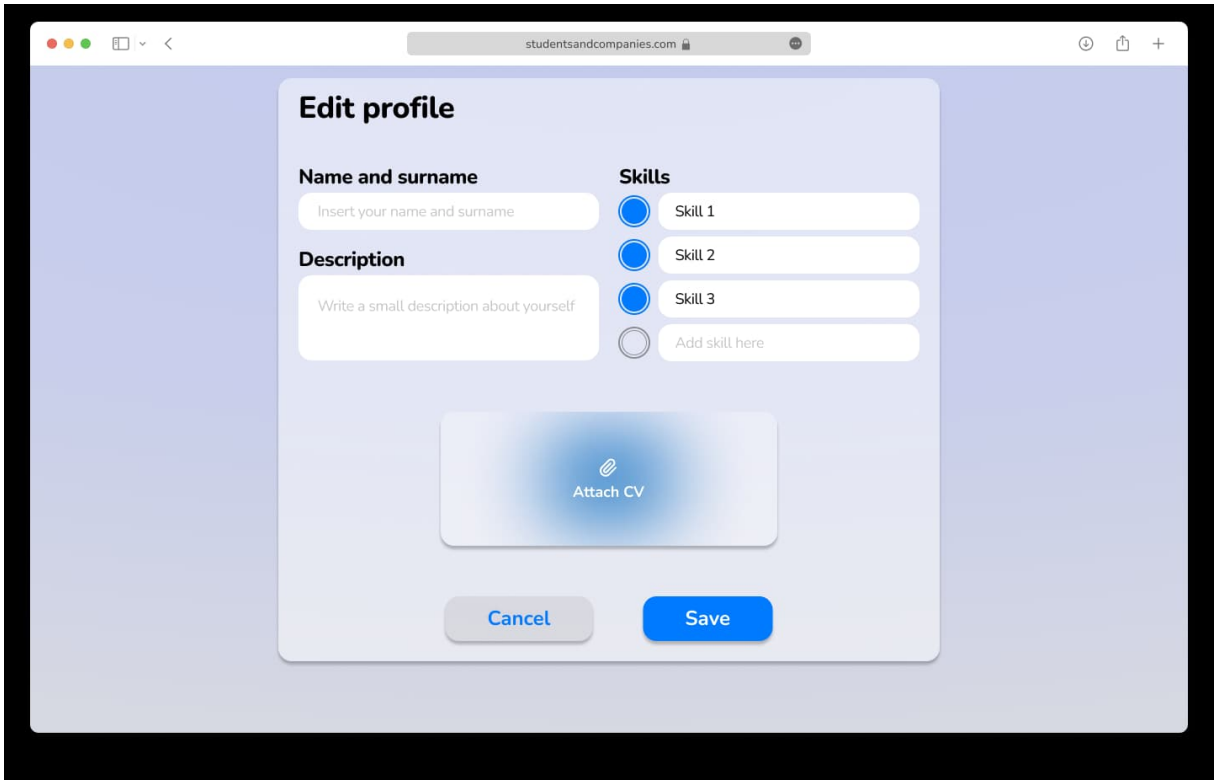
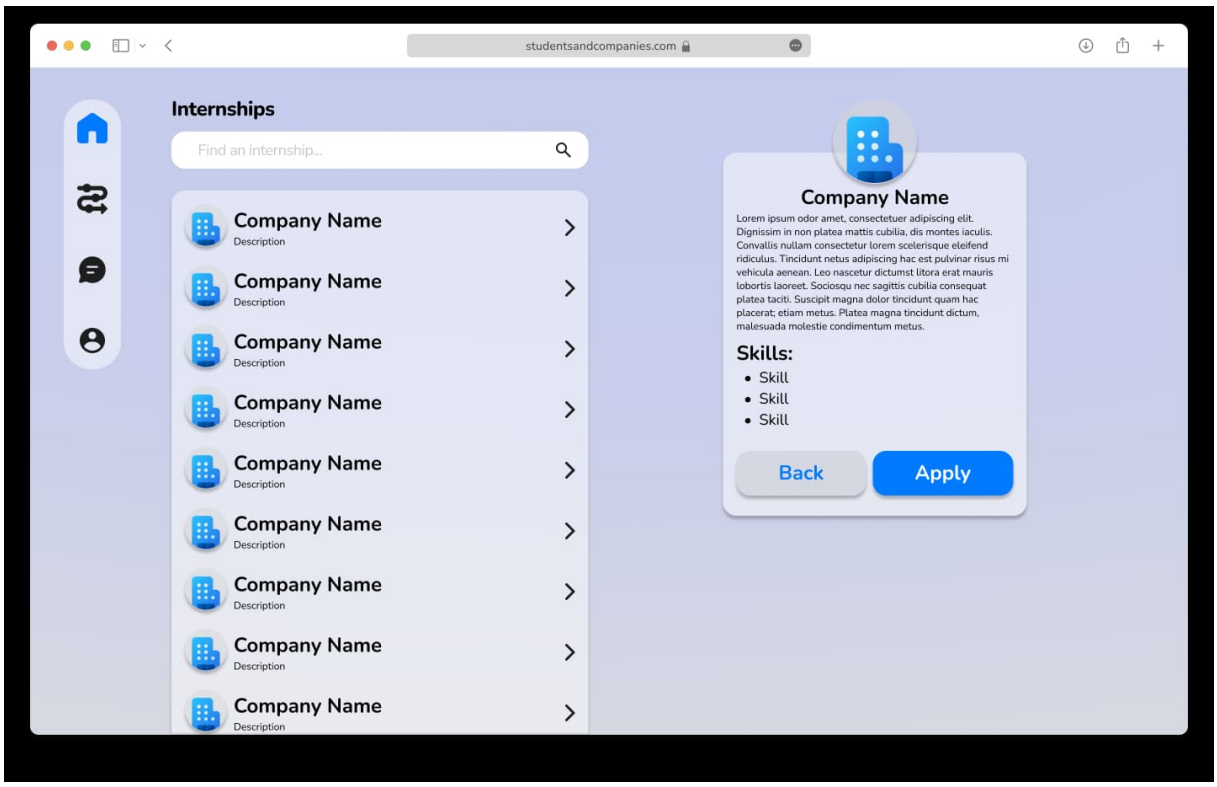


Figure 5: Search Internship



Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Figure 6: Create Questionnaire

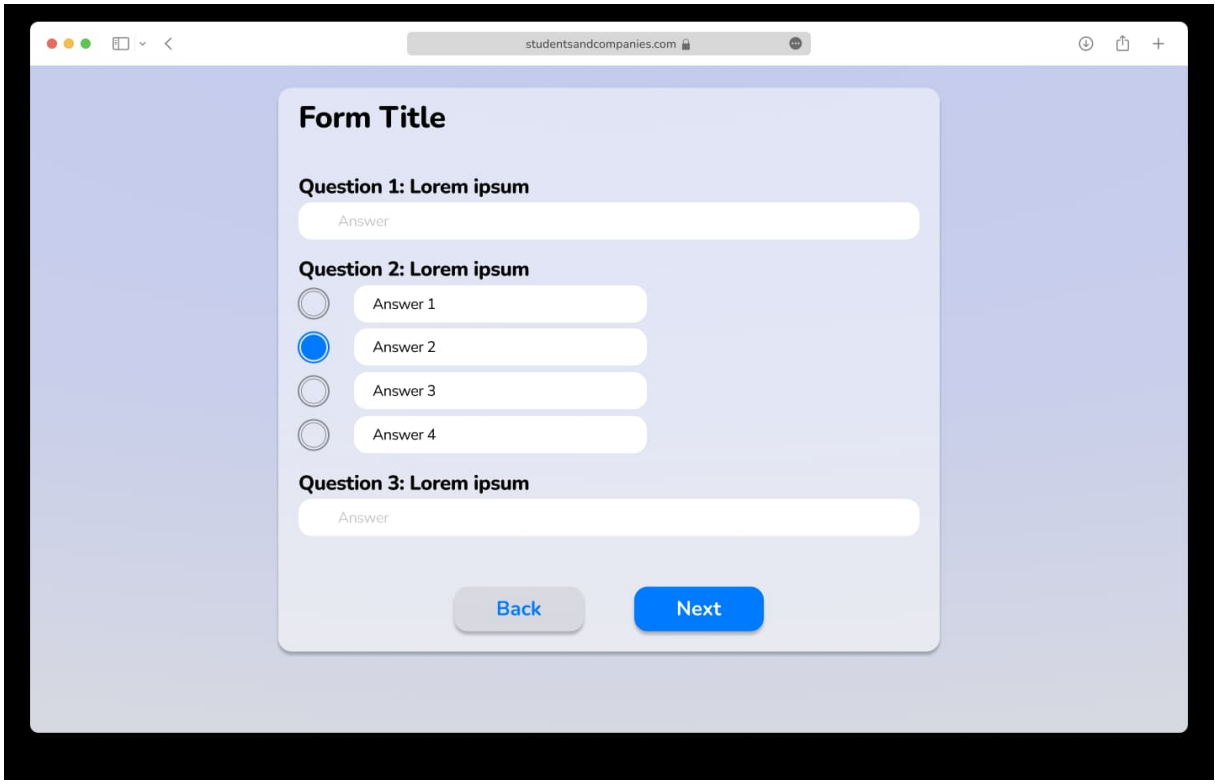
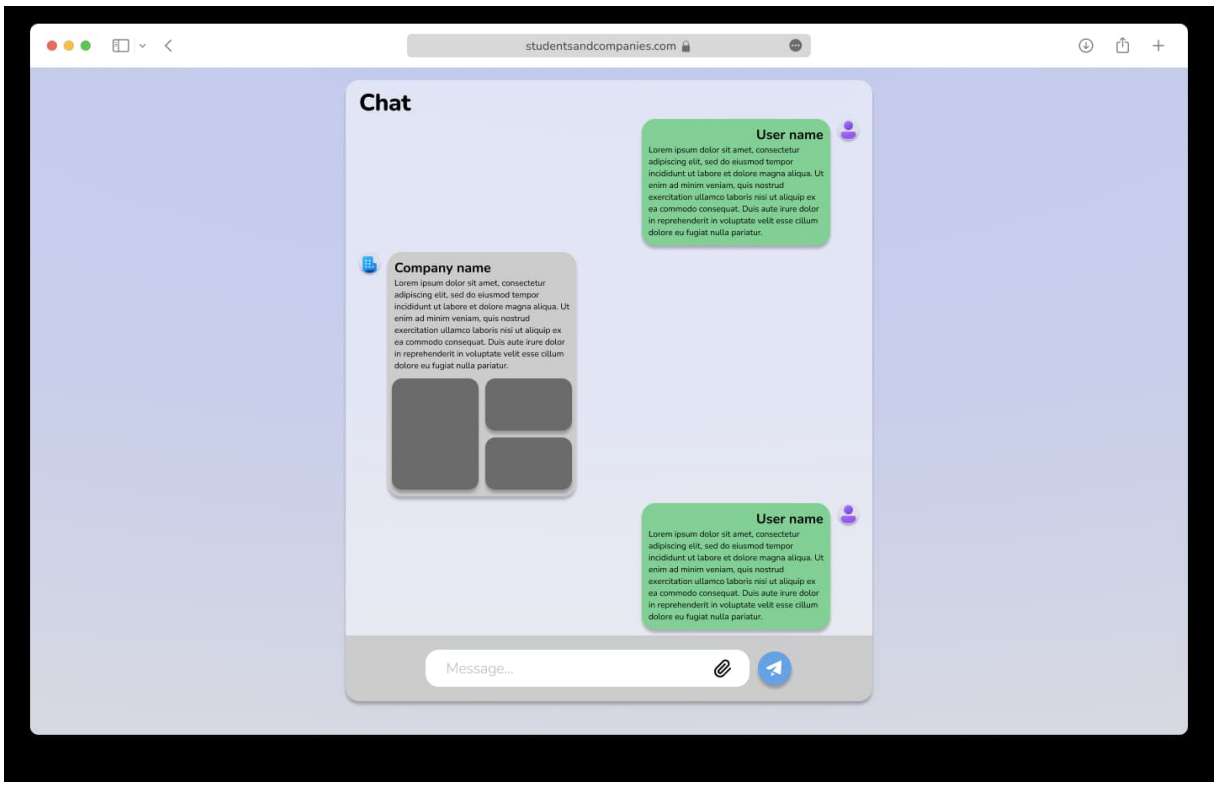


Figure 7: Discussion



Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Figure 8: Upload Internship

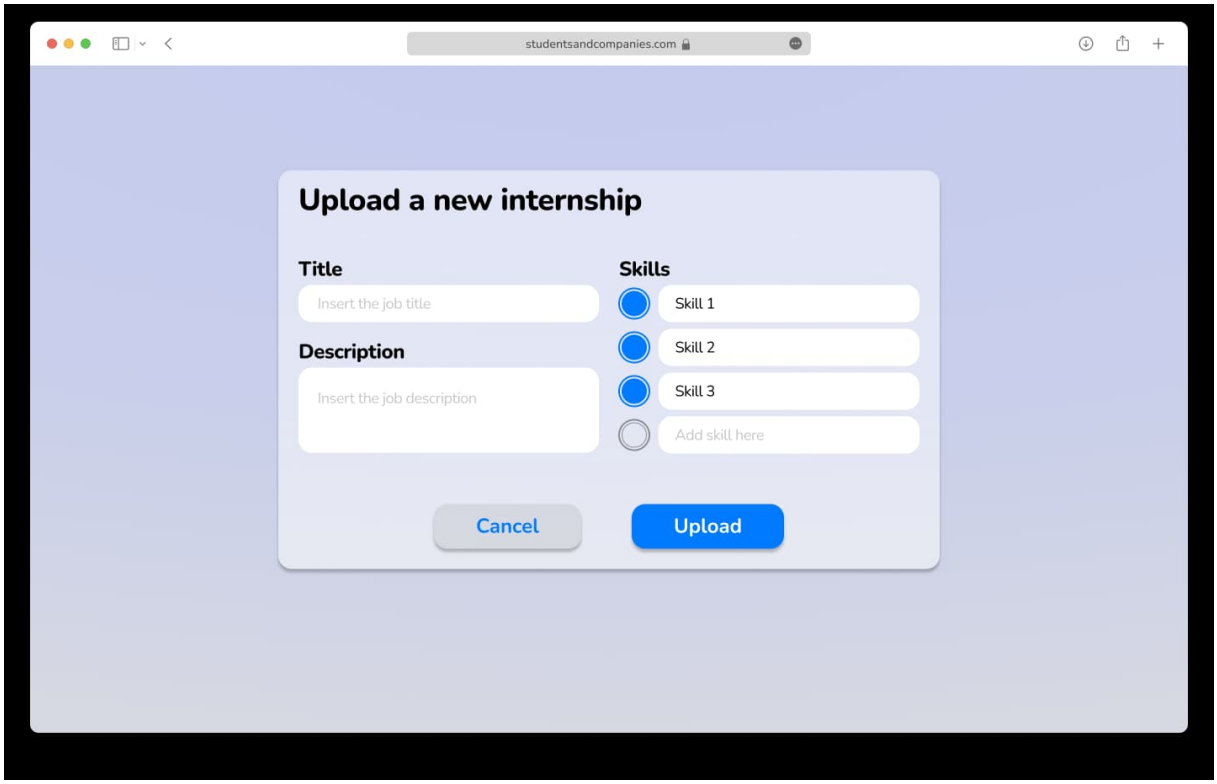
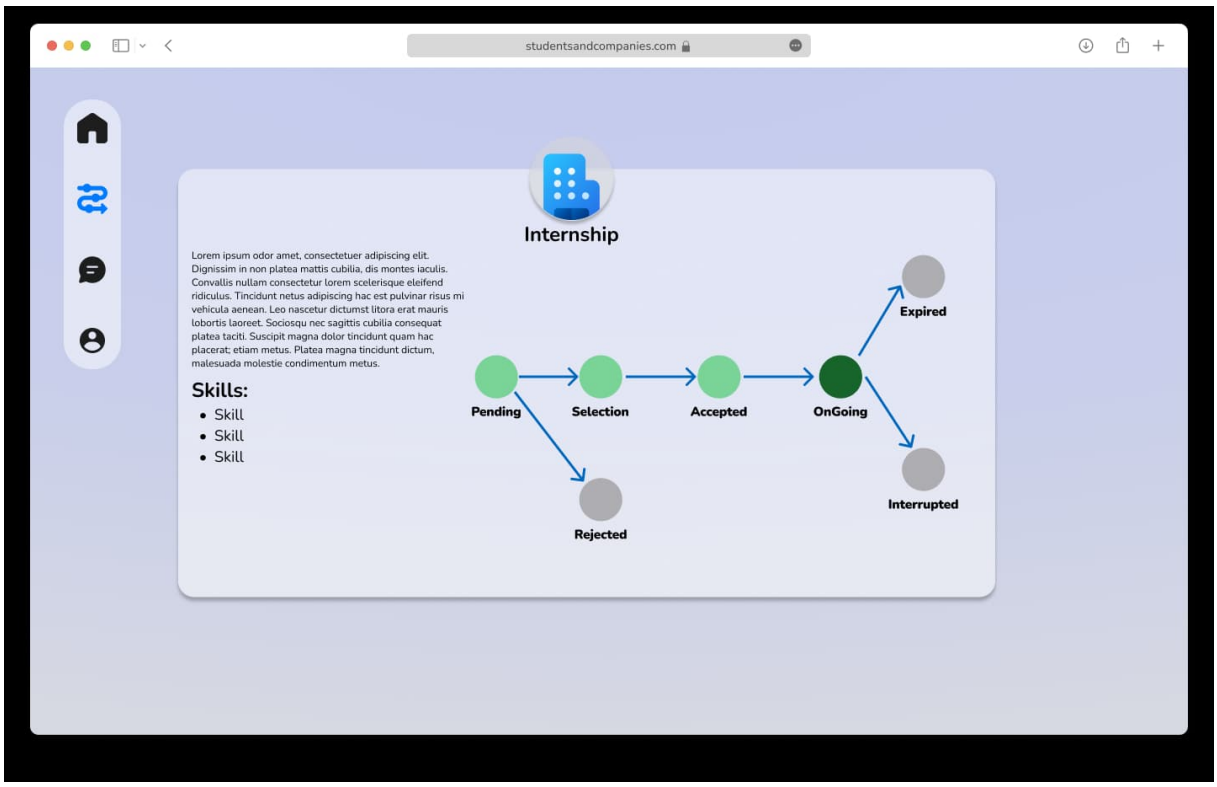


Figure 9: Monitor Status



Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

3.1.2 Hardware Interfaces

No hardware interfaces are provided, being S&C just a software system.

3.1.3 Software Interfaces

In order to work the system needs some software interfaces.

- **SheerID API**: in order to verify the students identity and his university, used also for user login.

3.1.4 Communication Interfaces

The system, which is entirely web-based, relies on HTTPS protocol.

3.2 Functional Requirements

3.2.1 Student

Login

[R1] The system must allow students to log-in through their academic email and password, if they provide them correctly.

[R2] Only students with an academic credential provided by a registered university can log-in.

Edit profile

[R3] The system must allow the student to edit his profile info, remove the CV and add a new version of the CV.

[R4] The student can edit only his profile.

Internship recommendation

[R5] The system must send an email to the student when an internship that might interest him becomes available.

Search an internship

[R6] The system must provide to the student all internships satisfying search criteria.

[R7] The system must provide to the student only internships satisfying search criteria.

Apply for an internship

[R8] The system must register the student's application for the internship.

[R9] The system must register only applications of students who have submitted an application for the internship.

Answer interview questions

[R10] The student must be able to answer interview questionnaires.

[R11] The student can answer interview questionnaires regarding only internships he's applied for and accepted by the involved company.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Consulting CV suggestion

[R12] The student must be able to view suggestions on how to improve his CV.

[R13] The student can view suggestions regarding only his CV.

3.2.2 Company

Registration and login

[R14] The system must allow companies to register to S&C providing mandatory data (name, email, VAT number) and selecting a password.

[R15] The system must allow registered companies to log-in through their email and password, if they are provided correctly.

Upload an internship

[R16] The system must allow the company to fill an internship creation form.

[R17] The system must allow the company to select the internship type.

[R18] The system must allow the company to select the internship duration.

[R19] The system must allow the company to select if the internship is paid or not.

[R20] The system must allow the company to select the skills it is searching for in the candidates.

Student recommendation

[R21] The system must send an email notifying the company when a student with a CV corresponding to their needs becomes available.

Interview students

[R22] The company must be able to provide structured questionnaires to students.

[R23] The company can provide structured questionnaires only to students who have applied to an internship uploaded by the company itself.

[R24] The company must be able to view answers to questionnaires it created.

Create contract

[R25] The company must be able to register a contract regarding an internship and referring to the assigned student.

[R26] The company can register contracts regarding only internships uploaded by the company itself.

[R27] The company can register contracts referring only to students that have applied for the indicated internship.

3.2.3 University

Login

[R28] The system must allow registered universities to log-in through credentials provided by S&C, if they are provided correctly.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Terminate an internship

[R29] The system must allow universities to terminate an internship involving their students.

3.2.4 Students and Companies

Leave a feedback

[R30] The student must be able to post a feedback.

[R31] The company must be able to post a feedback.

[R32] Only students that have participated in an internship can leave feedback for it.

[R33] Only companies that have participated in an internship can leave feedback for it.

[R34] Every student can leave at most one feedback per internship.

[R35] Every company can leave at most one feedback per internship.

Open discussion

[R36] The student must be able to open a discussion with the company he's interning with and the university he's enrolled at.

[R37] The company must be able to open a discussion with the students they are collaborating with and universities those students are enrolled at.

[R38] The student can open a discussion only with the company he's interning with and the university he's enrolled at.

[R39] The company can open a discussion only with students they are collaborating with and universities.

3.2.5 Everyone

Send and receive messages within a discussion

[R40] The student can send and receive messages only from a discussion he is participating in.

[R41] The company can send and receive messages only from a discussion it is participating in.

[R42] The university can send and receive messages only from a discussion it is participating in.

Monitor the status of the internship

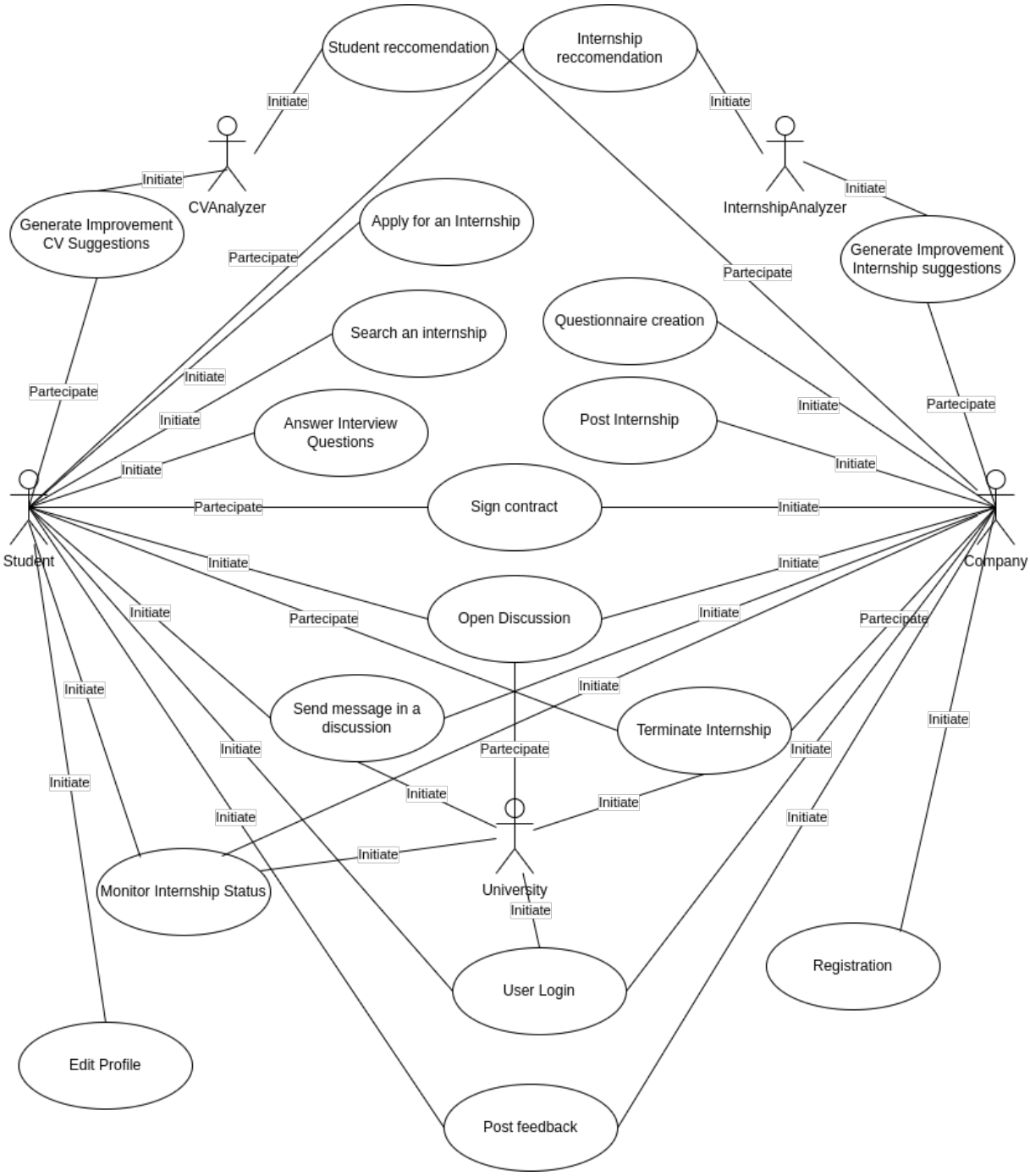
[R43] The student must be able to monitor the status of the internships in which he has applied.

[R44] The company must be able to monitor the status of the internships it has published.

[R45] The university must be able to monitor the status of the internship in which its students have applied.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

3.2.6 Use Cases Diagram



Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

3.2.7 Use Cases

[UC - 1] Company Registration

Name	Company registration
Actors	•Company
Entry condition	The company has clicked the register button in the login view.
Event flow	1.The user presses the register button. 2.The system shows the registration form. 3.The user compiles the form. 4.The user presses the submit button. 5.The platform shows the dashboard.
Exit condition	The company has successfully registered to the platform.
Exception	•There are missing data. •There already exists a user with those credentials. •Data format is not valid.

[UC - 2] Company Login

Name	Company Login
Actors	•Company
Entry condition	The company has opened the S&C platform.
Event flow	1.The company fills the form. 2.The company clicks the submit button. 3.The platform shows the dashboard.
Exit condition	The company has successfully logged into the platform
Exception	There does not exist any company with the submitted credentials.

[UC - 3] University Login

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Name	University Login
Actors	•University
Entry condition	The university has opened the S&C platform.
Event flow	1.The university fills the form. 2.The university clicks the submit button. 3.The platform shows the dashboard.
Exit condition	The university has successfully logged into the platform.
Exception	There does not exist any university with the submitted credentials.

[UC - 4] Student Login

Name	Student Login
Actors	•Student •SheerID
Entry condition	The student has opened the S&C platform.
Event flow	1.The student fills the form. 2.The student clicks the submit button. 3.The platform shows the dashboard.
Exit condition	The student has successfully logged into the platform.
Exception	There does not exist any student with the submitted credentials.

[UC - 5] Post Internship

Name	Post Internship
Actors	•Company
Entry condition	The company is logged to the S&C platform.
Event flow	1.The company clicks the upload internship button. 2.The company fills the form with the information about the internship: title, description, requested skills and the interview questionnaire.
Exit condition	The internship is successfully uploaded on the platform.
Exception	

[UC - 6] Post Feedback

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Name	Post Feedback
Actors	•Student •Company
Entry condition	The user is logged in and on the page of an internship he was involved in.
Event flow	1.The user clicks on the post feedback button. 2.The user fills the form with a rating and, optionally, a comment. 3.The user clicks on the submit button.
Exit condition	The feedback is successfully uploaded on the platform.
Exception	A required field was not filled.

[UC - 7] Search an Internship

Name	Search an Internship
Actors	•Student
Entry condition	The student is logged to the S&C platform.
Event flow	1.The student clicks the search icon. 2.The student selects any search criteria or filters. 3.The system displays the internships of the student that meet the selected filters, if any.
Exit condition	Internships are displayed to the student.
Exception	

[UC - 8] Apply for an Internship

Name	Apply for an Internship
Actors	•Student
Entry condition	The student is logged to the S&C platform.
Event flow	1.The student clicks on an internship post. 2.The student clicks the "apply for the internship" button.
Exit condition	The student has applied successfully for the internship.
Exception	

[UC - 9] Questionnaire Creation

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Name	Questionnaire Creations
Actors	•Company
Entry condition	The company is logged to the S&C platform.
Event flow	1.The company clicks the create questionnaire button. 2.The company fills the form with the questions to be inserted into the questionnaire. 3.The company clicks the submit button.
Exit condition	The questionnaire is successfully sent to the all the applicants.
Exception	

[UC - 10] Answer Questionnaire Interview

Name	Answer Questionnaire Interview
Actors	•Student
Entry condition	The student is logged to the S&C platform and there exists a contract between both parties.
Event flow	1.The student clicks on the answer questions button. 2.The system shows the questions to the student, 3.The student answer the questions. 4.The student clicks the submit button.
Exit condition	The questionnaire is successfully submitted and sent to the company.
Exception	

[UC - 11] Sign Contract

Name	Sign Contract
Actors	•Company
Entry condition	The company is logged to the S&C platform.
Event flow	1.The company selects the internship and the applicant to begin working with. 2.The company clicks the create contract button. 3.The company uploads the contract.
Exit condition	The student can start his internship.
Exception	Invalid contract format.

[UC - 12] Open Discussion

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Name	Open Discussion
Actors	•Student •Company
Entry condition	The user is logged to the S&C platform and there exists a contract between both parties.
Event flow	1.The user clicks on the open discussion button. 2.The user insert a title for the discussion. 3.The user clicks the submit button.
Exit condition	The discussion is successfully created.
Exception	

[UC - 13] Send Message in a Discussion

Name	Send Message in a Discussion
Actors	•User
Entry condition	The user must be logged to the S&C platform and be involved in an internship discussion.
Event flow	1.The user clicks on the internship. 2.The user clicks on the discussion. 3.The user writes a message. 4.The user clicks the send button. 5.The system sends the updated discussion view.
Exit condition	The message is sent and all the involved users can now see it, if logged in, in the discussion section.
Exception	

[UC - 14] Terminate Internship

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Name	Terminate Internship
Actors	•Student •Company •University
Entry condition	The university is logged to the S&C platform and there exists an internship contract between the student and the company.
Event flow	1.The university clicks on the internship. 2.The university clicks the "terminate internship" button. 3.A notification is sent to the involved student. 4.A notification is sent to the involved company. 5.A notification is sent to the university.
Exit condition	The internship is now terminated.
Exception	

[UC - 15] Generate Improvement CV Suggestion

Name	Generate Improvement CV Suggestion
Actors	•Student.
Entry condition	A CV/internship has been added/updated.
Event flow	1.The system analyzes existing CVs and internships. 2.The system generates, if possible, a suggestion. 3.The system sends to the students an email with suggestions on how to improve the CV.
Exit condition	The student has received a suggestion on how to improve his CV.
Exception	

[UC - 16] Generate Internship Improvement Suggestion

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Name	Generate Improvement Internship Suggestions
Actors	•Company.
Entry condition	A CV/internship has been added/updated.
Event flow	1.The system analyzes existing CVs and internships. 2.The system generates, if possible, a suggestion. 3.The system sends to the company an email with suggestions on how to improve the internship.
Exit condition	The company has received a suggestion on how to improve their internship.
Exception	

[UC - 17] Student Recommendation

Name	Student Recommendation
Actors	•Company
Entry condition	A CV has been added/updated.
Event flow	1.The system analyzes the CV. 2.If the system finds out that the CV might be interesting to a company, it sends an email to the company notifying the availability,
Exit condition	The company has received a notification about the availability of a new CV.
Exception	

[UC - 18] Internship Recommendation

Name	Internship Recommendation
Actors	•Student
Entry condition	An internship has been added.
Event flow	1.The system analyzes the new internship. 2.If the system finds that the new internship might interest a student, it sends an email to that student notifying the availability.
Exit condition	The company has received a notification about the availability of a new CV.
Exception	

[UC - 19] Monitor Internship Status

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

Name	Monitor Internship Status
Actors	•User
Entry condition	The user is either a student, a company or a university involved directly in an active internship.
Event flow	1.The user clicks on the internship to monitor. 2.The system shows the internship status and the eventual discussions.
Exit condition	The system has shown the internship dashboard with info.
Exception	

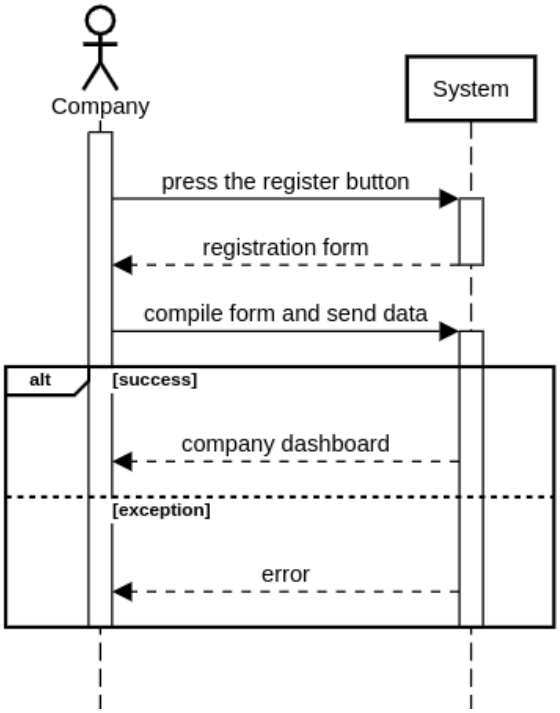
[UC - 20] Edit Profile

Name	Edit Profile
Actors	•Student
Entry condition	The student is logged to the S&C platform.
Event flow	1.The student clicks on the "edit profile" button. 2.The student selects the fields to modify, this includes the possibility to select the CV field. 3.The student changes the data in the selected fields. If the CV field was selected, the student uploads the new version of the CV. 4.The student clicks on the save button.
Exit condition	The system registers the new data
Exception	The student clicks the "cancel" button. Old data recorded in the system is preserved.

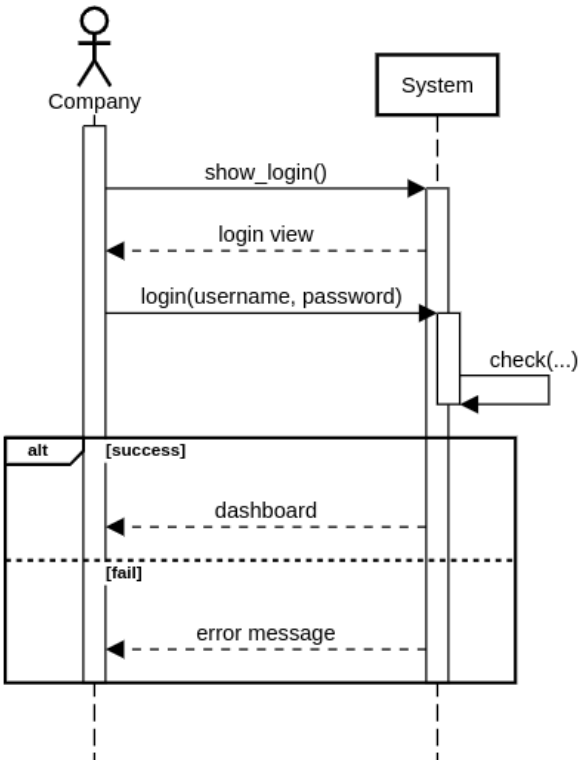
Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

3.2.8 Sequence Diagrams

[UC - 1] Company Registration

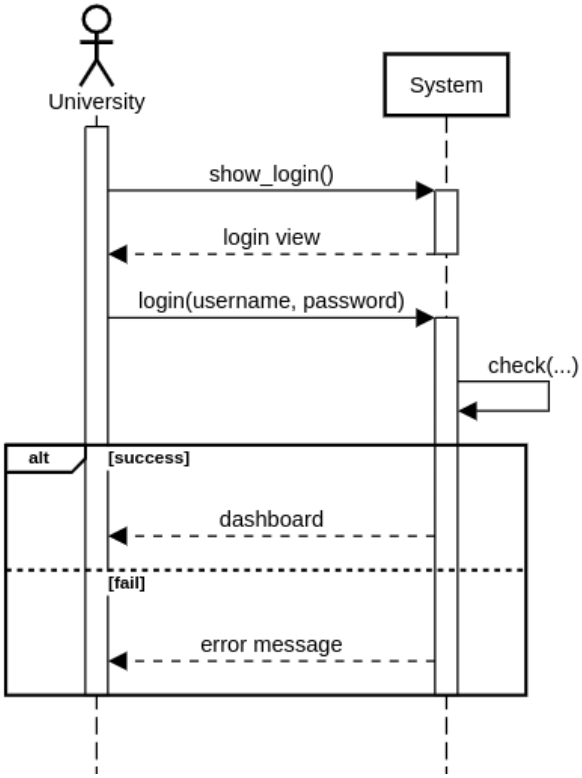


[UC - 2] Company Login

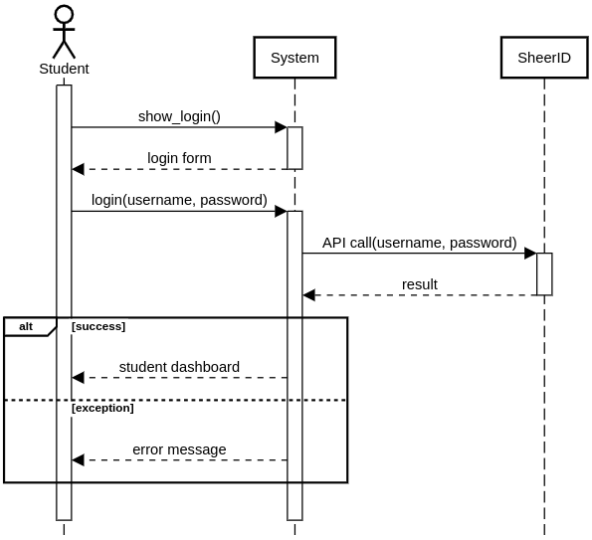


[UC - 3] University Login

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

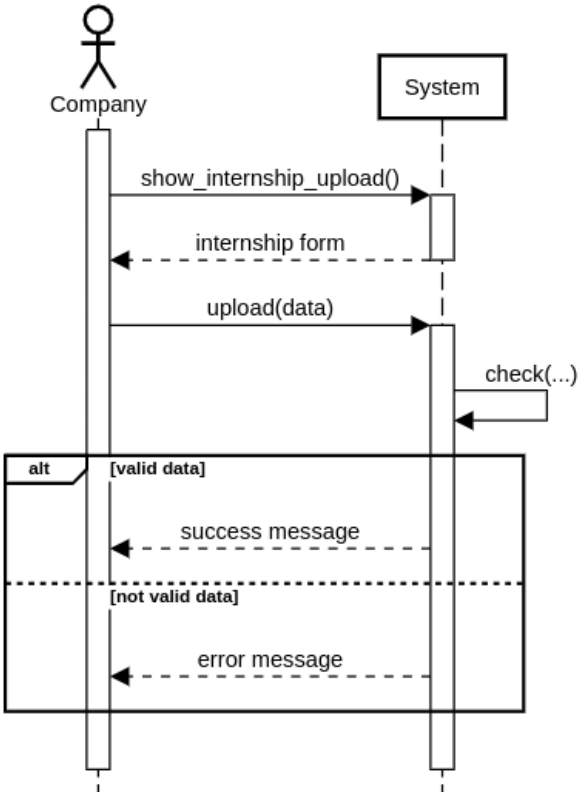


[UC - 4] Student Login

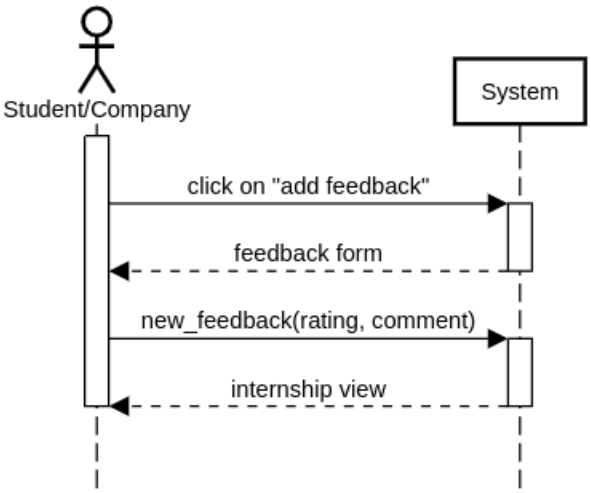


[UC - 5] Post Internship

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

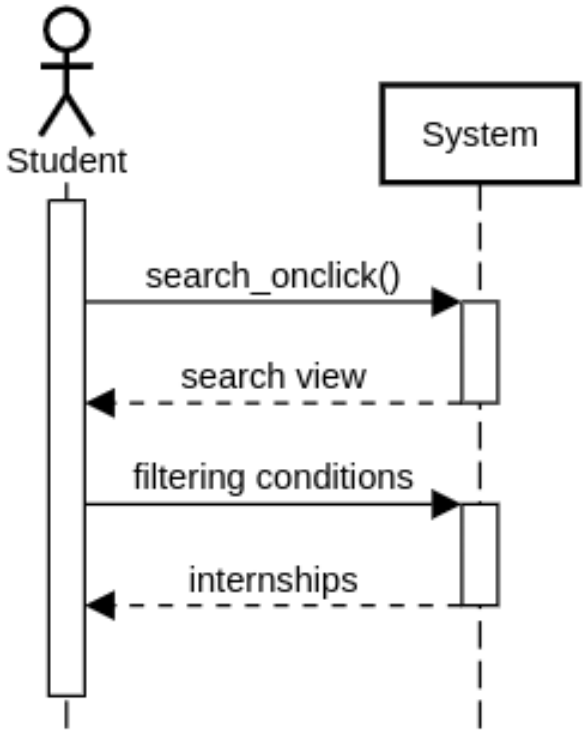


[UC - 6] Post Feedback

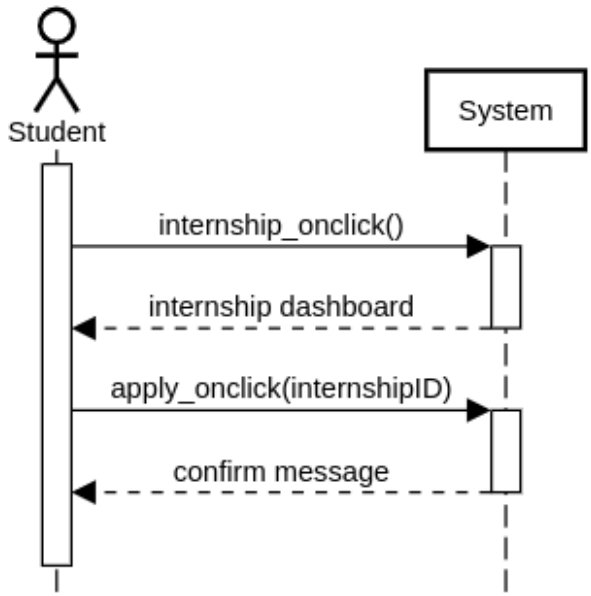


[UC - 7] Search an Internship

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

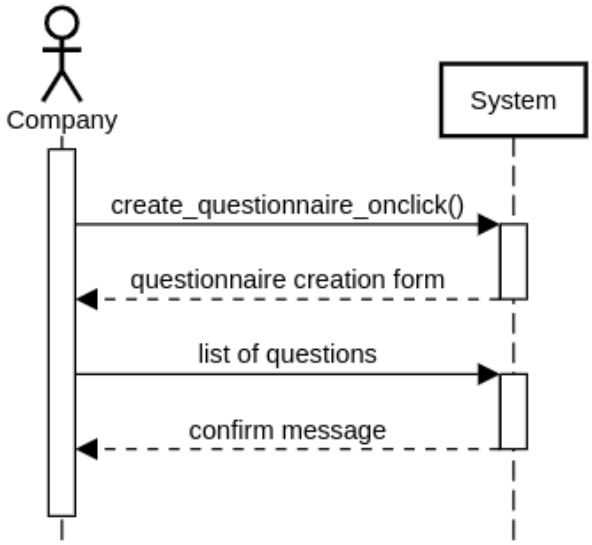


[UC - 8] Apply for an Internship

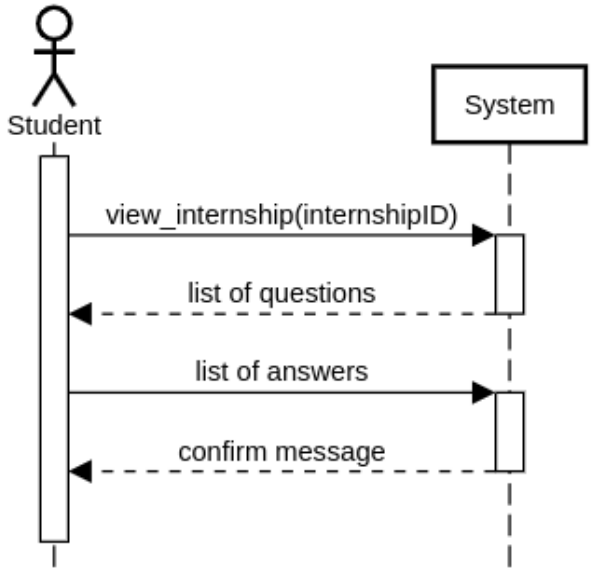


[UC - 9] Questionnaire Creation

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

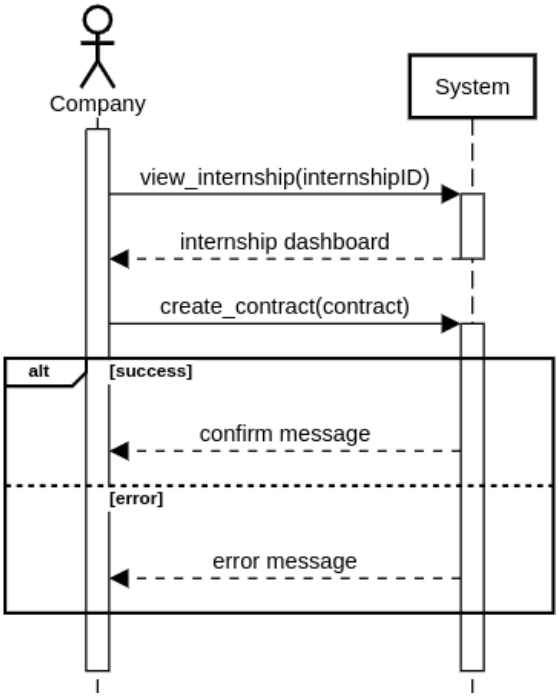


[UC - 10] Answer Questionnaire Interview

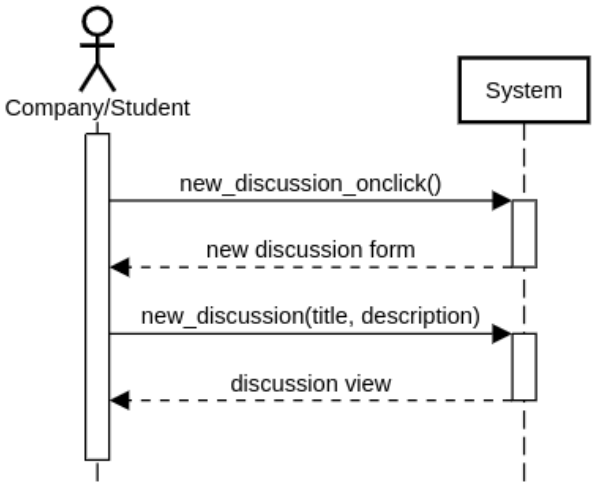


[UC - 11] Sign Contract

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

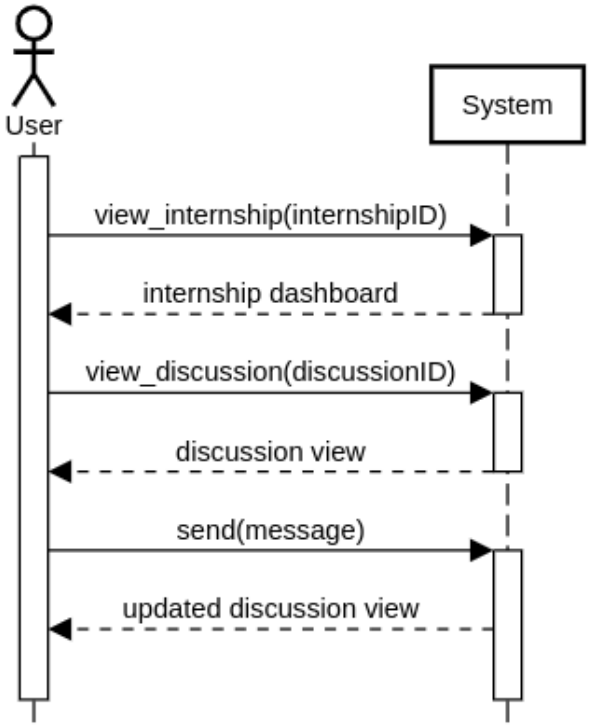


[UC - 12] Open Discussion

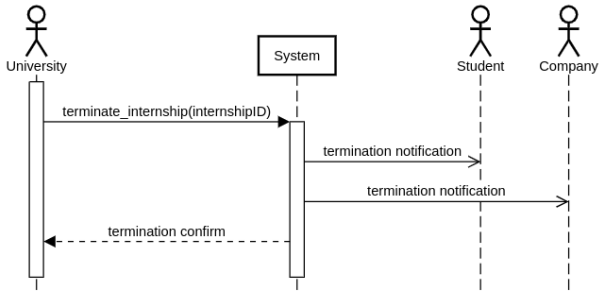


[UC - 13] Send Message in a Discussion

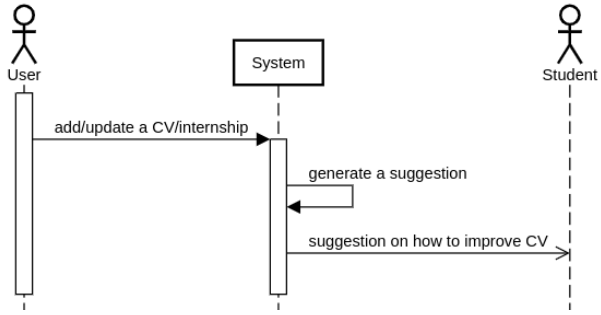
Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved



[UC - 14] Terminate Internship

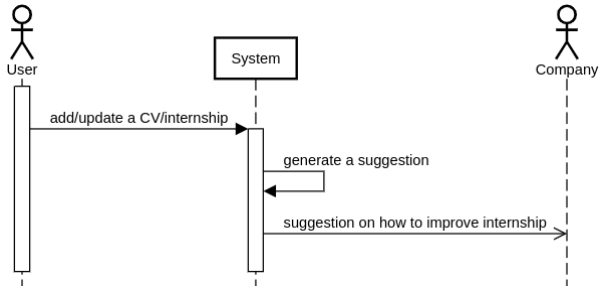


[UC - 15] Generate CV Improvement Suggestion

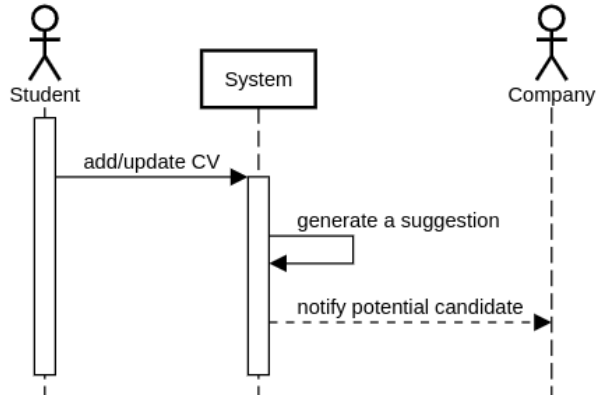


[UC - 16] Generate Internship Improvement Suggestion

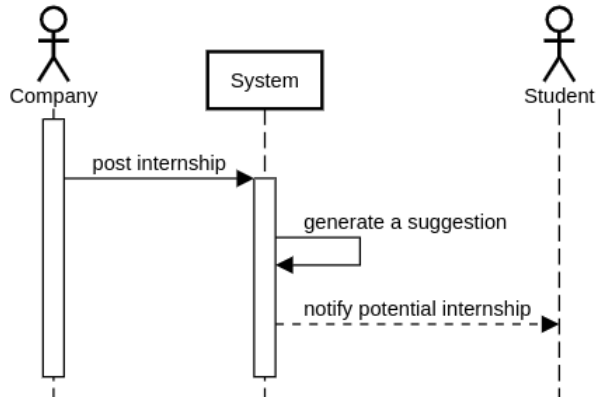
Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved



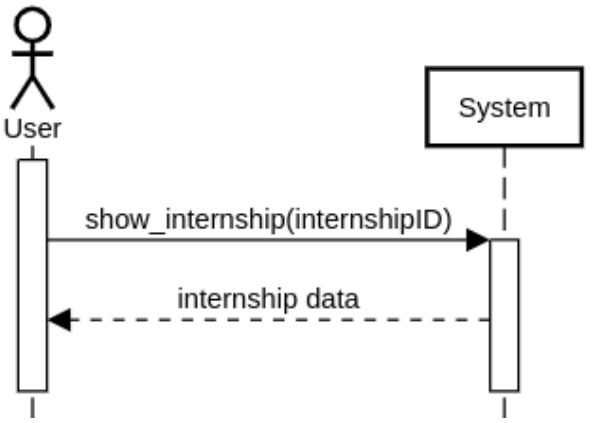
[UC - 17] Student Recommendation



[UC - 18] Internship Recommendation

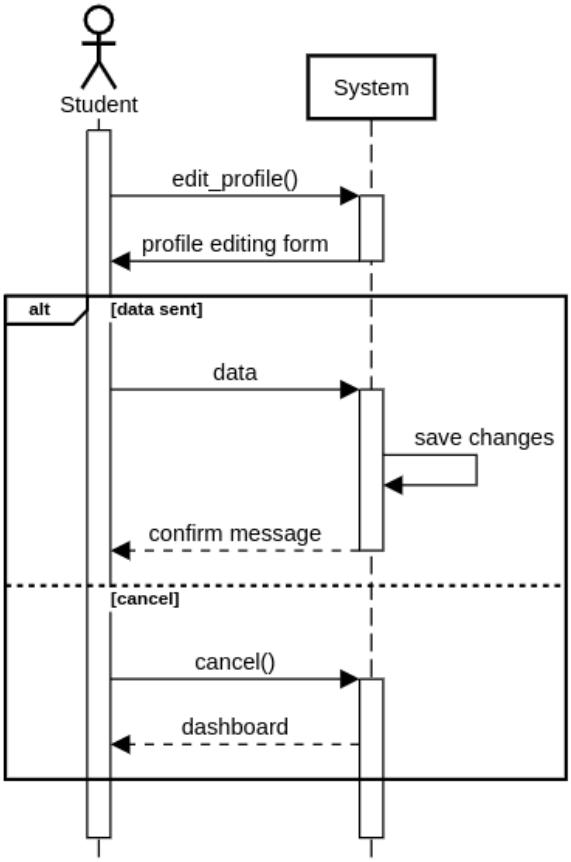


[UC - 19] Monitor Internship Status



Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[UC - 20] Edit Profile



Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

3.2.9 Requirement Mapping

[G1] Students must be able to search for internships posted by companies	
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password.	[D1] Students create their CVs
[R] The system must allow registered companies to log-in through their email and password, if they are provided correctly.	[D2] Companies properly insert informations about internships
[R] The system must allow the company to fill an internship creation form.	[D6] Students have an academic credential.
[R] The system must allow the company to select the internship type.	[D7] University is available on the SheerID API.
[R] The system must allow the company to select the internship duration.	[D8] Students are enrolled in a university registered on SheerID.
[R] The system must allow the company to select if the internship is paid or not.	[D9] Users who register as a company have a valid VAT number.
[R] The system must allow the company to select the skills it is searching for in the candidates.	
[R] The system must allow students to log-in through their academic email and password, if they provide them correctly.	
[R] Only students with an academic credential provided by a registered university can log-in.	
[R] The system must provide to the student all internships satisfying search criteria.	
[R] The system must provide to the student only internships satisfying search criteria.	

[G2] Companies must be able to upload their internships offers	
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password.	[D2] Companies properly insert information about internships.
[R] The system must allow registered companies to log-in through their email and password, if they are provided correctly.	[D9] Users who register as a company have a valid VAT number.
[R] The system must allow the company to fill an internship creation form.	
[R] The system must allow the company to select the internship type.	
[R] The system must allow the company to select the internship duration.	
[R] The system must allow the company to select if the internship is paid or not.	
[R] The system must allow the company to select the skills it is searching for in the candidates.	

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G3] Students must be notified when an internship that could interest them becomes available.	
[R] The system must allow students to log-in through their academic email and password, if they provide them correctly. [R] Only students with an academic credential provided by a registered university can log-in. [R] The system must provide to the student all internships satisfying search criteria. [R] The system must provide to the student only internships satisfying search criteria.	[D2] Companies properly insert information about internships. [D4] Notifications must arrive to the target users. [D6] Students have an academic credential. [D7] University is available on the SheerID API. [D8] Students are enrolled in a university registered on SheerID.

[G4] Companies must be notified about available students with CVs and skills matching their internship requirements.	
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password. [R] The system must allow registered companies to log-in through their email and password, if they are provided correctly. [R] The system must allow the company to fill an internship creation form. [R] The system must allow the company to select the internship type. [R] The system must allow the company to select the internship duration. [R] The system must allow the company to select if the internship is paid or not. [R] The system must allow the company to select the skills it is searching for in the candidates. [R] The system must send an email notifying the company when a student with a CV corresponding to their needs becomes available.	[D1] Students create their CVs. [D2] Companies properly insert information about internships. [D4] Notifications must arrive to the target users. [D9] Users who register as a company have a valid VAT number.

[G5] Students must be able to apply for internships of interest.	
[R] The system must allow students to log-in through their academic email and password, if they provide them correctly. [R] Only students with an academic credential provided by a registered university can log-in. [R] The system must provide to the student all internships satisfying search criteria. [R] The system must provide to the student only internships satisfying search criteria.	[D1] Students create their CVs. [D2] Companies properly insert information about internships. [D6] Students have an academic credential. [D7] University is available on the SheerID API. [D8] Students are enrolled in a university registered on SheerID.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G6] Companies must be able to accept applicants to their internship.	
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password. [R] The system must allow registered companies to log-in through their email and password, if they are provided correctly. [R] The company must be able to register a contract regarding an internship and referring to the assigned student. [R] The company can register contracts regarding only internships uploaded by the company itself. [R] The company can register contracts referring only to students that have applied for the indicated internship.	[D2] Companies properly insert information about internships.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G7] Companies must be supported in conducting a selection process for applicants, including the collection of information about the students using questionnaires and the management of interviews.	
[R] The system must allow students to log-in through their academic email and password, if they provide them correctly.	[D1] Students create their CVs.
[R] Only students with an academic credential provided by a registered university can log-in.	[D2] Companies properly insert information about internships.
[R] The student must be able to answer interview questionnaires.	[D4] Notifications must arrive to the target users.
[R] The student can answer interview questionnaires regarding only internships he's applied for and accepted by the involved company.	[D6] Students have an academic credential.
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password.	[D7] University is available on the SheerID API.
[R] The system must allow registered companies to log-in through their email and password, if they are provided correctly.	[D8] Students are enrolled in a university registered on SheerID.
[R] The system must allow the company to fill an internship creation form.	[D9] Users who register as a company have a valid VAT number.
[R] The system must allow the company to select the internship type.	
[R] The system must allow the company to select the internship duration.	
[R] The system must allow the company to select if the internship is paid or not.	
[R] The system must allow the company to select the skills it is searching for in the candidates.	
[R] The company must be able to provide structured questionnaires to students.	
[R] The company can provide structured questionnaires only to students who've applied to an internship uploaded by the company itself.	
[R] The company must be able to view answers to questionnaires it created.	

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G8] Students want to receive suggestions on how to make their CVs more appealing.	
[R] The system must allow students to log-in through their academic email and password, if they provide them correctly.	[D1] Students create their CVs.
[R] Only students with an academic credential provided by a registered university can log-in.	[D2] Companies properly insert information about internships.
[R] The system must provide to the student all internships satisfying search criteria.	[D4] Notifications must arrive to the target users.
[R] The system must provide to the student only internships satisfying search criteria.	[D6] Students have an academic credential.
[R] The student must be able to view suggestions on how to improve his CV.	[D7] University is available on the SheerID API.
[R] The student can view suggestions regarding only his CV.	[D8] Students are enrolled in a university registered on SheerID.
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password.	[D9] Users who register as a company have a valid VAT number.
[R] The system must allow registered companies to log-in through their email and password, if they are provided correctly.	
[R] The system must allow the company to fill an internship creation form.	
[R] The system must allow the company to select the internship type.	
[R] The system must allow the company to select the internship duration.	
[R] The system must allow the company to select if the internship is paid or not.	
[R] The system must allow the company to select the skills it is searching for in the candidates.	

[G9] Companies want to receive suggestions on how to make their project descriptions more appealing.	
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password.	[D2] Companies properly insert information about internships.
[R] The system must allow registered companies to log-in through their email and password, if they are provided correctly.	[D4] Notifications must arrive to the target users.
[R] The system must allow the company to fill an internship creation form.	[D9] Users who register as a company have a valid VAT number.
[R] The system must allow the company to select the internship type.	
[R] The system must allow the company to select the internship duration.	
[R] The system must allow the company to select if the internship is paid or not.	
[R] The system must allow the company to select the skills it is searching for in the candidates.	

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G10.A] Students must be able to keep track and monitor the execution and the outcomes of the match-making process.	
[R] The system must allow students to log-in through their academic email and password, if they provide them correctly.	[D1] Students create their CVs.
[R] Only students with an academic credential provided by a registered university can log-in.	[D2] Companies properly insert information about internships.
[R] The system must provide to the student all internships satisfying search criteria.	[D4] Notifications must arrive to the target users.
[R] The system must provide to the student only internships satisfying search criteria.	[D6] Students have an academic credential.
[R] The system must register the student's application for the internship.	[D7] University is available on the SheerID API.
[R] The system must register only applications of students who have submitted an application for the internship.	[D8] Students are enrolled in a university registered on SheerID.
[R] The student must be able to monitor the status of the internships in which he has applied.	[D9] Users who register as a company have a valid VAT number.
[R] The company must be able to monitor the status of the internships it has published.	
[R] The university must be able to monitor the status of the internship in which its students have applied.	

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G10.B] Companies must be able to keep track and monitor the execution and the outcomes of the matchmaking process.	
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password.	[D1] Students create their CVs.
[R] The system must allow registered companies to log-in through their email and password, if they are provided correctly.	[D2] Companies properly insert information about internships.
[R] The system must allow the company to fill an internship creation form.	[D4] Notifications must arrive to the target users.
[R] The system must allow the company to select the internship type.	[D6] Students have an academic credential.
[R] The system must allow the company to select the internship duration.	[D7] University is available on the SheerID API.
[R] The system must allow the company to select if the internship is paid or not.	[D8] Students are enrolled in a university registered on SheerID.
[R] The system must allow the company to select the skills it is searching for in the candidates.	[D9] Users who register as a company have a valid VAT number.
[R] The system must send an email notifying the company when a student with a CV corresponding to their needs becomes available.	
[R] The student must be able to monitor the status of the internships in which he has applied.	
[R] The company must be able to monitor the status of the internships it has published.	
[R] The university must be able to monitor the status of the internship in which its students have applied.	

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G10.C] Universities must be able to monitor the status of the internship in which its students have applied.	
[R] The system must register the student's application for the internship. [R] The system must register only applications of students who have submitted an application for the internship. [R] The system must allow the company to fill an internship creation form. [R] The system must allow the company to select the internship type. [R] The system must allow the company to select the internship duration. [R] The system must allow the company to select if the internship is paid or not. [R] The system must allow the company to select the skills it is searching for in the candidates. [R] The system must allow registered universities to log-in through credentials provided by S&C, if they are provided correctly. [R] The student must be able to monitor the status of the internships in which he has applied. [R] The company must be able to monitor the status of the internships it has published. [R] The university must be able to monitor the status of the internship in which its students have applied.	[D1] Students create their CVs. [D2] Companies properly insert information about internships. [D3] Universities interfere with the contract between students and companies when needed. [D4] Notifications must arrive to the target users. [D6] Students have an academic credential. [D7] University is available on the SheerID API. [D8] Students are enrolled in a university registered on SheerID. [D9] Users who register as a company have a valid VAT number.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G11] The interested parties of an internship (students, companies and universities) must be able to communicate with each other about info and eventual problems regarding the ongoing internship.	
[R] The system must allow students to log-in through their academic email and password, if they provide them correctly.	[D3] Universities interfere with the contract between students and companies when needed.
[R] Only students with an academic credential provided by a registered university can log-in.	[D4] Notifications must arrive to the target users.
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password.	[D6] Students have an academic credential.
[R] The system must allow registered companies to log-in through their email and password, if they are provided correctly.	[D7] University is available on the SheerID API.
[R] The system must allow registered universities to log-in through credentials provided by S&C, if they are provided correctly.	[D8] Students are enrolled in a university registered on SheerID.
[R] The student must be able to open a discussion with the company he's interning with and the university he's enrolled at.	
[R] The company must be able to open a discussion with the students they are collaborating with and universities those students are enrolled at.	
[R] The student can open a discussion only with the company he's interning with and the university he's enrolled at.	
[R] The company can open a discussion only with students they are collaborating with and universities.	
[R] The student can send and receive messages only from a discussion he is participating in.	
[R] The company can send and receive messages only from a discussion it is participating in.	
[R] The university can send and receive messages only from a discussion it is participating in.	

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G12.A] Companies must be able to file complaints about students interning with them.	
[R] The system must allow companies to register to S&C providing mandatory data (name, email, P.IVA) and selecting a password. [R] The system must allow registered companies to log-in through their email and password, if they are provided correctly. [R] The company must be able to open a discussion with the students they are collaborating with and universities those students are enrolled at. [R] The company can open a discussion only with students they are collaborating with and universities. [R] The student can send and receive messages only from a discussion he is participating in. [R] The company can send and receive messages only from a discussion it is participating in. [R] The university can send and receive messages only from a discussion it is participating in.	[D3] Universities interfere with the contract between students and companies when needed. [D4] Notifications must arrive to the target users. [D9] Users who register as a company have a valid VAT number.

[G12.B] Students must be able to file complaints about the companies where they are interning.	
[R] The system must allow students to log-in through their academic email and password, if they provide them correctly. [R] Only students with an academic credential provided by a registered university can log-in. [R] The student must be able to open a discussion with the company he's interning with and the university he's enrolled at. [R] The student can open a discussion only with the company he's interning with and the university he's enrolled at. [R] The student can send and receive messages only from a discussion he is participating in. [R] The company can send and receive messages only from a discussion it is participating in. [R] The university can send and receive messages only from a discussion it is participating in.	[D3] Universities interfere with the contract between students and companies when needed. [D4] Notifications must arrive to the target users. [D6] Students have an academic credential. [D7] University is available on the SheerID API. [D8] Students are enrolled in a university registered on SheerID.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

[G13] Universities must be able to review complaints about their students and the companies where their students are interning.	
[R] The system must allow registered universities to log-in through credentials provided by S&C, if they are provided correctly.	[D3] Universities interfere with the contract between students and companies when needed.
[R] The company must be able to open a discussion with the students they are collaborating with and universities those students are enrolled at.	[D4] Notifications must arrive to the target users.
[R] The company can open a discussion only with students they are collaborating with and universities.	[D6] Students have an academic credential.
[R] The company can only send and receive messages from a discussion it is participating in.	[D7] University is available on the SheerID API.
[R] The university can only send and receive messages from a discussion it is participating in.	[D8] Students are enrolled in a university registered on SheerID.

[G14] Universities must be able to interrupt internships between their students and a company.	
[R] The system must allow registered universities to log-in through credentials provided by S&C, if they are provided correctly.	[D3] Universities interfere with the contract between students and companies when needed.
[R] The system must allow universities to terminate an internship involving their students	[D4] Notifications must arrive to the target users.
[R] The student must be able to open a discussion with the company he's interning with and the university he's enrolled at.	[D6] Students have an academic credential.
[R] The company must be able to open a discussion with the students they are collaborating with and universities those students are enrolled at.	[D7] University is available on the SheerID API.
[R] The student can open a discussion only with the company he's interning with and the university he's enrolled at.	[D8] Students are enrolled in a university registered on SheerID.
[R] The company can open a discussion only with students they are collaborating with and universities.	

3.3 Performance Requirements

Performance requirements are not particularly critical for the system, but it is anyway desirable that all requests sent to the server are answered within 1 second, to assure a good user experience.
The server infrastructure will be designed to be scalable so that it will be possible to adapt it to the increment of users when the app diffusion will increase.

3.4 Design Constraints

3.4.1 Standards Compliance

The CKB platform pays a great attention for what concerns users privacy cause of this the CKB project is in compliance with the General Data Protection Regulation (GPDR) a regulation in EU law on data protection and privacy for all individuals within the European Union (EU) and the European Economic Area (EEA). Moreover, the platform has to use the international format of date and time to adequate to the newer standards.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

3.4.2 Hardware and Software Limitations

There are not significant limitations since users can access the system using every modern browser

3.5 Software System Attributes

3.5.1 Availability

The availability is not a critical requirement, but the system has to guarantee a 99% of uptime (max 3.65 days/year of downtime) to ensure that the users can normally use it.

3.5.2 Security

Security is a critical requirement for this system, considering the confidential information that is transmitted through it. It is assured by the use of the HTTPS protocol for all communications and by following the best security practices for the servers management, maintaining the data ciphered and assuring the access only to the authorized users.

3.5.3 Maintainability

The system will be realized following the best software engineering practices to ensure its maintainability and expandability in the future.

3.5.4 Portability

The system is a web application so it must be compatible with modern web browsers (Firefox, Google Chrome and so on) and feature a responsive interface to be used on a wide range of screen formats (smartphones, tablets, computers, etc).

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

4 Formal Analysis Using Alloy

In the following section the system is modeled with Alloy[1] to check that the contracts' execution flow is correct and that related actions such as opening a discussion or giving a feedback can be performed only in the intended phases.

```
// Signatures
sig User {
        name: one String
}

sig Student extends User {
        university: one University,
        skills: set String
}

sig Company extends User {
        internships: set Internship
}

sig University extends User {}

sig Internship {
        offeredBy: one Company,
        title: one String,
        skillsRequired: set String
} { this in offeredBy.internships }

sig Contract {
        internship: one Internship,
        student: one Student,
        var status: one ContractStatus
}

enum ContractStatus { Pending, Accepted, OnGoing, Rejected, Expired,
↪  Interrupted }

sig Discussion {
        owner: one User,
        contract: one Contract,
        messages: set DiscussionMessage
} { (owner not in University) and
        (owner = contract.student or owner = contract.student.university or
        ↪  owner = contract.internship.offeredBy) and
        (all m: DiscussionMessage | m in messages implies (m.sender =
        ↪  contract.student or m.sender = contract.student.university or
        ↪  m.sender = contract.internship.offeredBy))}

sig DiscussionMessage {
        sender: one User,
        content: one String
```

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

```
} { one d: Discussion | this in d.messages }

sig Feedback {
        contract: one Contract,
        author: one User,
        content: one String,
        rating: one Int
} { author in contract.student + contract.internship.offeredBy and rating >= 1
↪  and rating <= 5}

// Flow control
fact newIsPending {
        all c: Contract | c.status = Pending
}

// Possible flows
pred NormalFlow[c: Contract] { (c.status = Pending ; c.status = Accepted ;
↪  c.status = OnGoing ; always c.status = Expired) }
pred InterruptedFlow[c: Contract] { (c.status = Pending ; c.status = Accepted ;
↪  c.status = OnGoing ; always c.status = Interrupted) }
pred RejectedFlow[c: Contract] { (c.status = Pending ; always c.status =
↪  Rejected) }

// No other flows other that the ones listed
fact NoOtherFlows {
        no c: Contract |
                not c.RejectedFlow and
                not c.InterruptedFlow and
                not c.NormalFlow
}

// General facts

// Every User is a Student, Company or Univesity
fact AllUsersSubclassed { all u: User | u in Student + Company + University }

// Each Student can't apply twice to the same internship
fact CannotApplyTwiceToAnInternship {
        all c1, c2: Contract | (c1.student = c2.student and c1.internship =
        ↪  c2.internship) implies c1 = c2
}

// To interrupt a contract, a discussion must have been opened at least once
fact InterruptionNeedDiscussion {
    always (all c: Contract | c.status = Interrupted implies (some d:
    ↪  Discussion | d.contract = c))
}

// Can't give feedback if the contract has reached an end (the contract must
↪  have not been rejected)
```

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

```
fact FeedbackAllowedOnlyIfCompletion {
    always (all f: Feedback, c: Contract | c = f.contract implies eventually
    ↪  c.status in (Expired + Interrupted))
}

// Can't Open a discussion if the contract has been rejected
fact NoDiscussionIfRejectcion {
         always (all d: Discussion, c: Contract | c = d.contract implies
         ↪  c.status not in Rejected)
}

// Can't give multiple feedbacks to the same internship experience
fact NoMultipleFeedbacks{
    all f1, f2: Feedback |
        (f1.contract = f2.contract and f1.author = f2.author) implies f1 = f2
}

// Every internship is posted by a company
fact EveryInternshipHasACompany {
    all i: Internship | one c: Company | i in c.internships
}

// No multiple students working in the same internship at the same time
fact NoMultipleOngoingContractForSameIntership {
    always all c1,c2: Contract | (c1.status = OnGoing and c2.status = OnGoing
    ↪  and c1.internship = c2.internship) implies c1.student = c2.student
}

// To match students and companies there must be at least one matching skill
↪  between them
fact MatchingSkillsDuringContractCreation {
    all c: Contract |
        some c.student.skills & c.internship.skillsRequired
}

// Assertions

// Duplicate applications for the same combination of student and internship
↪  are not allowed
assert NoDuplicateApplications {
    all c1, c2: Contract |
        (c1.student = c2.student and c1.internship = c2.internship) implies c1
        ↪  = c2
}

check NoDuplicateApplications for 10

// DiscussionMessages can be sent only by the parties of the Contract or the
↪  University of the contracted Student
assert NoMessagesFromSenderOutsideContract{
```

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

```
    all d: Discussion, m: DiscussionMessage, c: Contract |
        (d.contract = c and m in d.messages) implies
                ((d.owner = c.student or d.owner = c.internship.offeredBy) and
                (m.sender = c.student or m.sender = c.student.university or
                ↪  m.sender = c.internship.offeredBy))
}

check NoMessagesFromSenderOutsideContract for 10

// Show
// Show with matching skills
pred ShowWithSkills {
    some u: University, s: Student, c: Company, i: Internship {
           (some con: Contract | con.NormalFlow)
        s.university = u
        s.name = "Pluto"
        s.skills = "java" + "c"
        i.skillsRequired = "java" + "python"
        i in c.internships
           i.title = "Programmer"
           u.name = "Polimi"
           c.name = "PoliGrammer"
    }
}

// Show normal flow of action
pred ShowSingle {
        some u: University, s: Student, c: Company, i: Internship {
        (some con: Contract | con.NormalFlow)
                s.university = u
                s.name = "Pluto"
               i in c.internships
    }
}

// Show interrupted contract
pred ShowInterrupted {
        some u: University, s: Student, c: Company, i: Internship {
        (some con: Contract | con.InterruptedFlow)
                s.university = u
                s.name = "Pluto"
               i in c.internships
    }
}

// General show (no required skills)
pred Show {
        some u: University, s: Student, c: Company, i: Internship {
                (some con: Contract | con.RejectedFlow)
                (some con: Contract | con.NormalFlow)
```

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

```
                (some con: Contract | con.InterruptedFlow)
                s.university = u
                  s.name = "Pluto"
               i in c.internships
    }
}

// Single
run ShowSingle for 4 but exactly 1 Contract, exactly 1 Student,exactly 1
↪  Company, exactly 1 University, exactly 1 Internship

// Status progression
run Show for 4 but exactly 3 Contract, exactly 2 Student, 3 Company, 2
↪  University, 4 Internship, exactly 1 Discussion, exactly 0 Feedback

// Feedback
run Show for 4 but exactly 1 Student, exactly 1 Company, 1 University, exactly
↪  4 Feedback

// Discussion
run ShowInterrupted for 4 but exactly 1 Contract, exactly 1 Student, exactly 1
↪  Company, exactly 1 University, exactly 1 Internship, exactly 2 Discussion,
↪  exactly 3 DiscussionMessage

// Skill matching
run ShowWithSkills for 4 but exactly 1 Contract, exactly 1 Student,exactly 1
↪  Company, exactly 1 University, exactly 1 Internship, exactly 0 Feedback,
↪  exactly 0 Discussion
```

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

4.1 Examples

An example of a contract's standard progression where it eventually expires without any issues.

```
run ShowSingle for 4 but exactly 1 Contract, exactly 1 Student,exactly 1 Company,
exactly 1 University, exactly 1 Internship
```

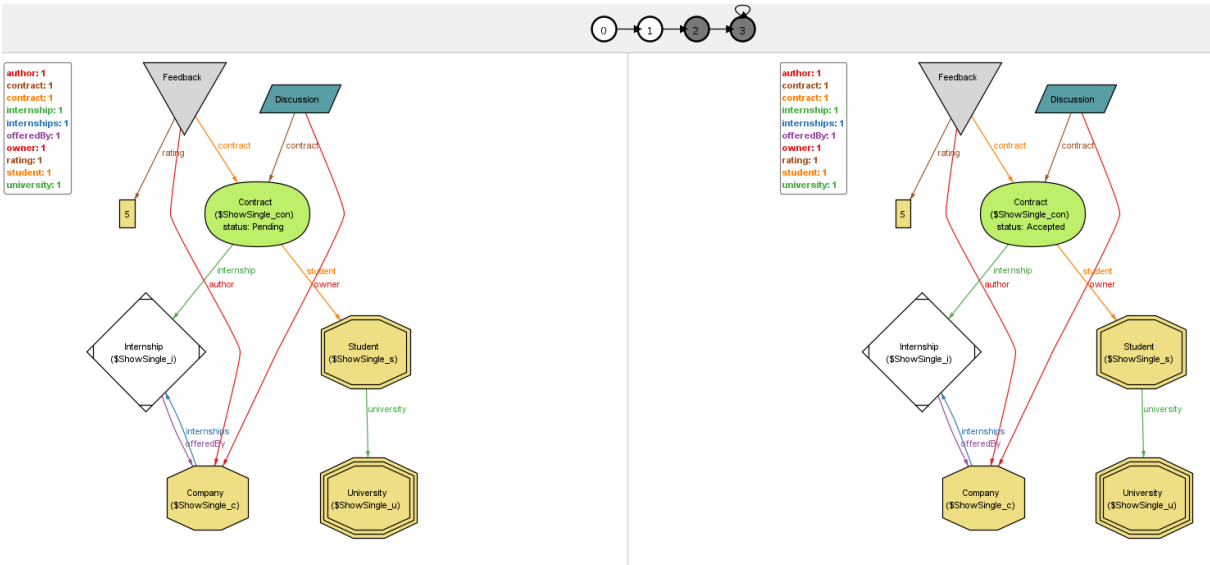


Figure 10: ShowSingle, first two steps

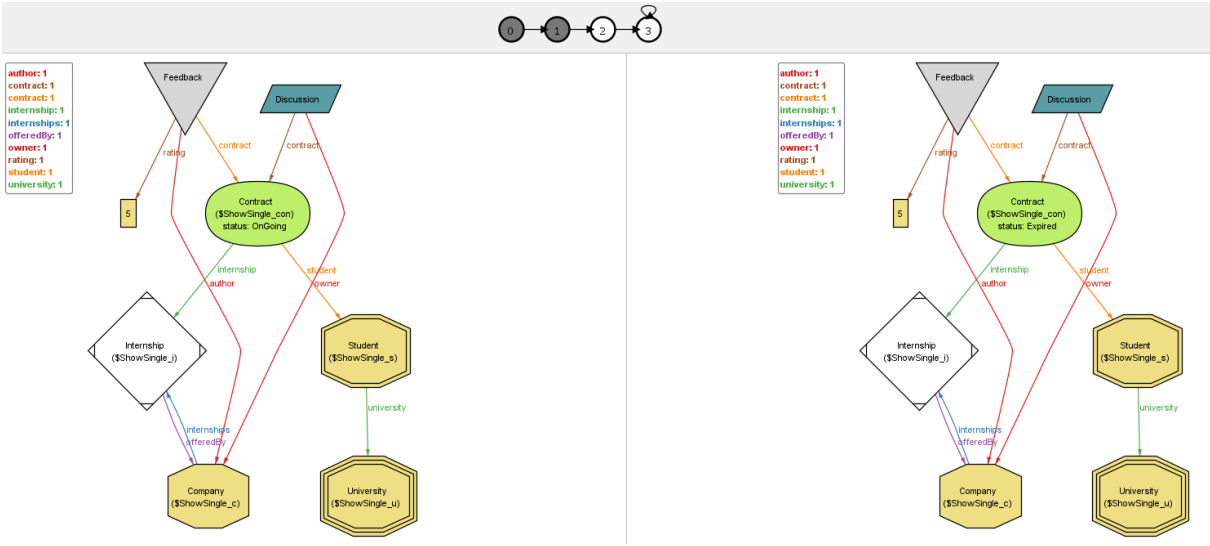


Figure 11: ShowSingle, first two steps

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

An example of the progression for every contract.

```
run Show for 4 but exactly 3 Contract, exactly 2 Student, 3 Company, 2 University,
4 Internship, exactly 1 Discussion, exactly 0 Feedback
```

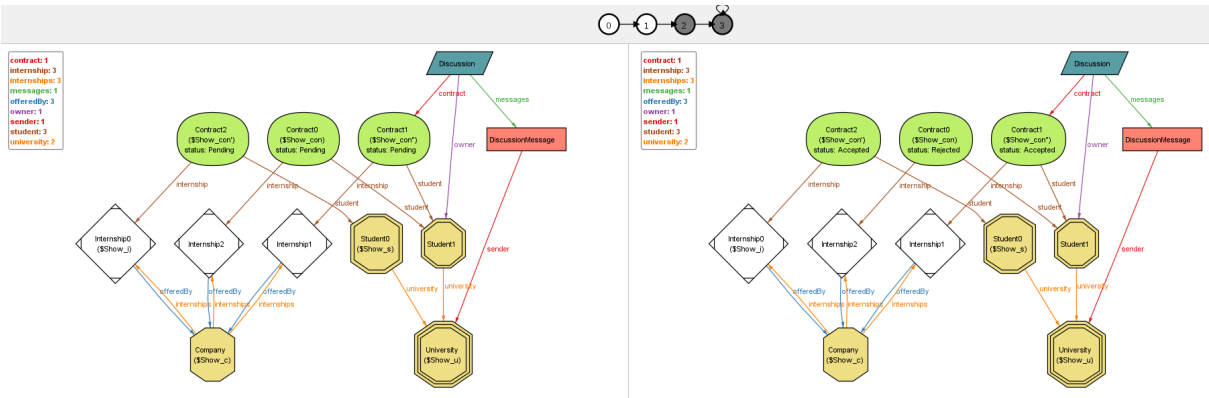


Figure 12: Show, first two steps

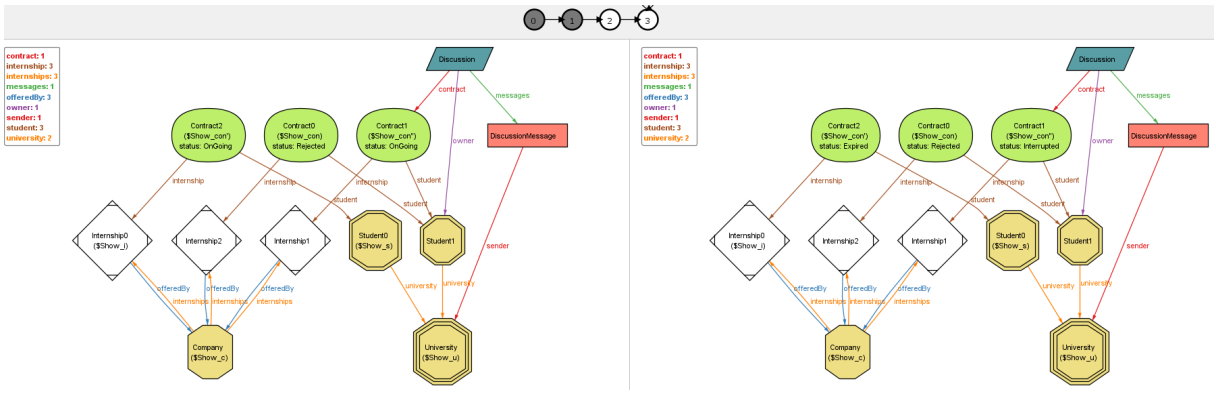


Figure 13: Show, last two steps

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

An example showing that every non-rejected contract can have feedback from both parties involved in the contract.

```
run Show for 4 but exactly 1 Student, exactly 1 Company, 1 University, exactly
4 Feedback
```

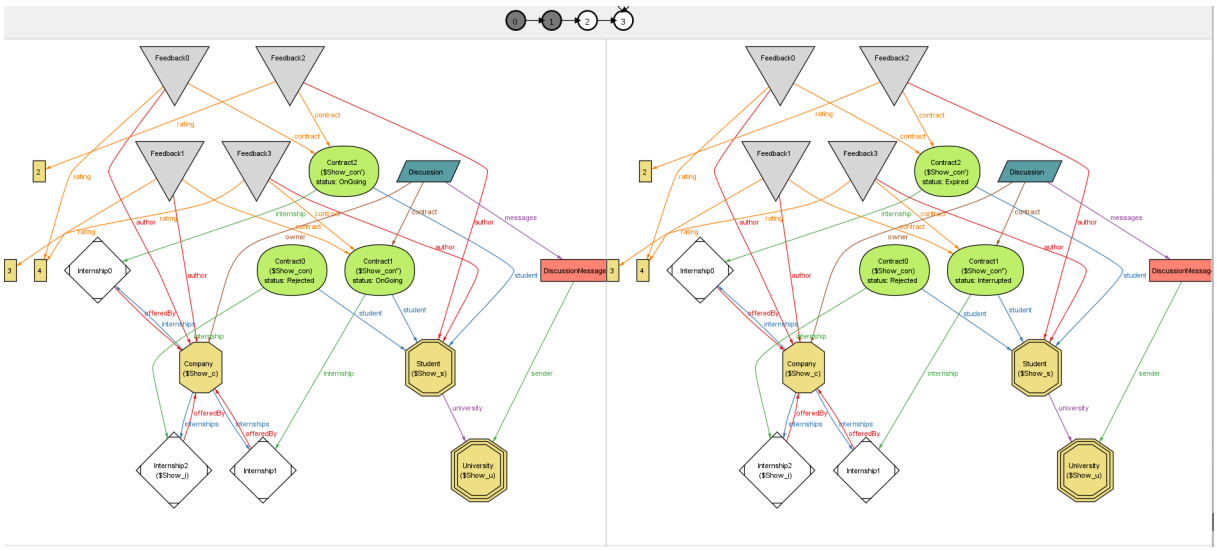


Figure 14: Show with Feedback, last two steps

An example demonstrating how the interruption of a contract works (at least one discussion must have been created beforehand).

```
run ShowInterrupted for 4 but exactly 1 Contract, exactly 1 Student, exactly 1
Company, exactly 1 University, exactly 1 Internship, exactly 2 Discussion, exactly
3 DiscussionMessage
```

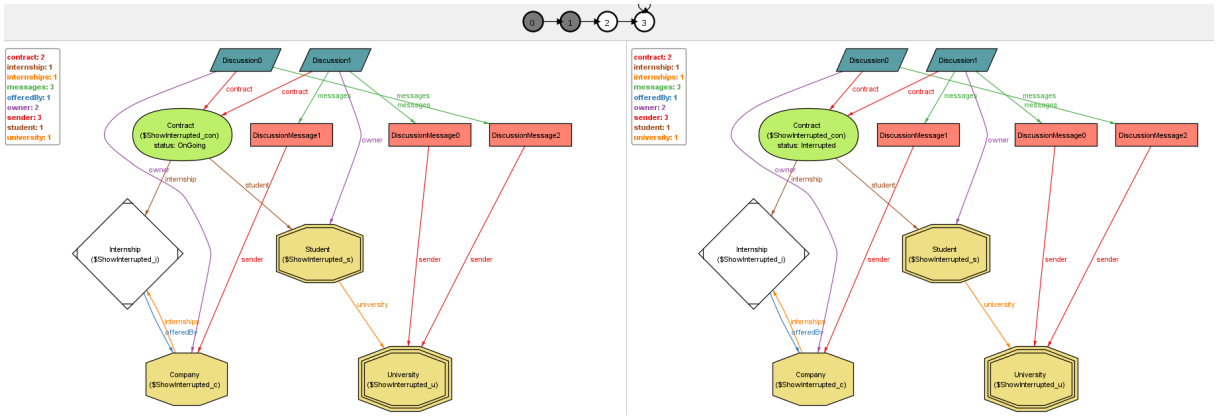


Figure 15: ShowInterrupted, last two steps

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

An example illustrating how skill matching works.

```
run ShowWithSkills for 4 but exactly 1 Contract, exactly 1 Student,exactly 1 Company,
exactly 1 University, exactly 1 Internship, exactly 0 Feedback, exactly 0 Discussion
```

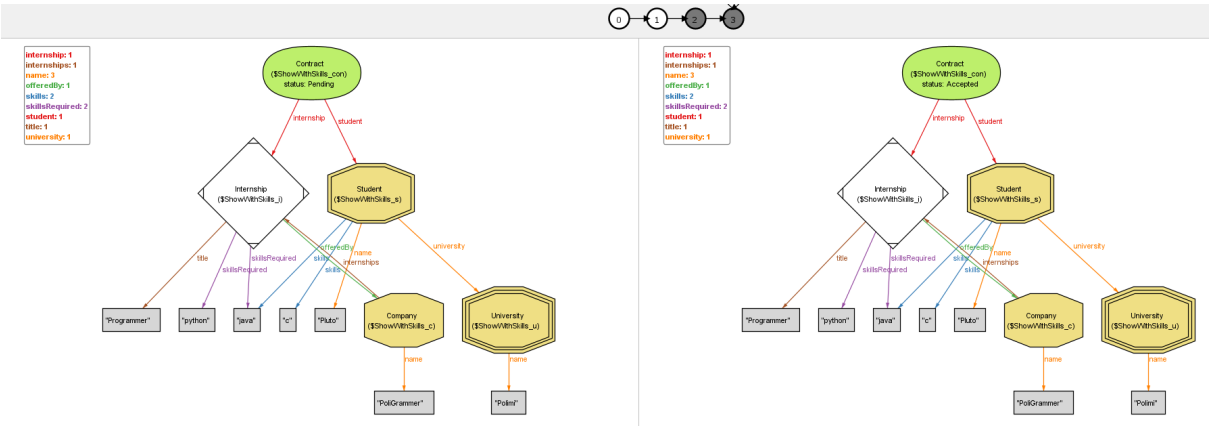


Figure 16: ShowWithSkills, first two steps

Checks

Every check terminated without finding counterexamples, showing that the checked are valid.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

5 Effort Spent

Bogdanovic Stefan

Section	Effort spent
1	8
2	13
3	20
4	3

Rodari Simone

Section	Effort spent
1	6
2	8
3	12
4	23

Pignataro Gabriele

Section	Effort spent
1	8
2	14
3	11
4	10

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved

References

[1] Alloy. Alloy documentation, 2023. URL: https://alloy.readthedocs.io.

[2] SheerID Inc. SheerID, 2024. URL: https://www.sheerid.com.

Copyright © 2024, Bogdanovic Stefan, Rodari Simone, Pignataro Gabirele – All rights reserved